

Five Facts About (Rental) Prices: Landlord Heterogeneity and the Dynamics of Shelter Inflation*

Sarah Baker and Caleb Wroblewski

September 21, 2026

Abstract

Housing is the largest component of the CPI, yet little is known about how landlords set contract rents. Using more than two decades of administrative new-tenant lease data from Berkeley, California, and the nationally representative American Housing Survey, we document five facts about rental price dynamics. First, new-tenant rents exhibit state-dependent downward nominal rigidity: although 13% of leases reset at exactly the previous tenant's rent, the missing mass of rent cuts arises only in expansions and largely disappears in downturns. Second, unlike many non-housing prices, both the frequency and size of rent changes vary strongly with market conditions. Third, rents display substantial seasonality that is comparable to the home-purchase market. Fourth, rents cluster at round numbers and just below them, consistent with coarse pricing, left-digit pricing, and misoptimization by landlords. This clustering intensifies when rent growth is unusually high or low, revealing a novel cognitive cost of inflation volatility: landlords price less precisely when nominal adjustments become more complex. Fifth, larger landlords use less coarse pricing and adjust rents more aggressively over the business cycle. We use these facts to estimate a hybrid model of rental price setting with time-dependent reset opportunities and costly re-optimization. The model implies that replacing small landlords with large landlords would raise average new-tenant rent inflation by 0.40 percentage points and increase its cyclical-ity. Our results provide new evidence on shelter price setting and highlight the role of behavioral frictions and landlord heterogeneity in inflation dynamics.

*We thank seminar and conference participants at the UEA Annual Meetings and the Federal Reserve Bank of Philadelphia, as well as Christian Hilber and Jón Steinsson for helpful comments. This paper previously circulated under the (abbreviated) title Five Facts About (Rental) Prices. Contact: sarah.baker@berkeley.edu, caleb.wroblewski@gmail.com

1 Introduction

Housing is the largest regular expense for most U.S. households, making shelter prices central to understanding household budgets and economic well-being. Shelter accounts for 35% of total expenditures in the Consumer Price Index (CPI), about the same as all goods combined. These costs also play a key role in shaping aggregate inflation dynamics over the business cycle. From 1990-2019, core (non-shelter) CPI remained stable, while the rental component covaried strongly with the business cycle (see Figure 1). Similarly, during the post-Covid inflation period, the non-shelter components of CPI peaked at around 6%, while rental price inflation peaked at nearly 9%.

Despite the central role of rental housing in shaping both aggregate inflation and household budgets, price-setting in rental markets remains poorly understood. Much of what we know about price-setting in goods markets comes from retail scanner data, which links transactions to clearly defined products. No equivalent dataset exists for rental housing. Recent studies have used advertised rents from multiple listing services or platforms like Craigslist, but these data cover only selected segments of the market, are limited to recent years, and track posted, rather than transacted, rents. The composition of listings also shifts over time, complicating inference about broader market dynamics.

We use a unique, comprehensive administrative registry of new-tenant rents in Berkeley, California, a city located in the San Francisco Bay Area and part of one of the nation’s most important rental markets. This dataset allows us to provide new evidence on rental price setting over the last 25 years. Our dataset allows us to follow contract rents for new tenants at the unit level. Our setting is ideal for several reasons. First, Berkeley has experienced significant variation in rental prices over the last few decades. Thus, we can study rental price setting in normal times, as well as in periods when rental prices are falling or rising rapidly. Our sample period includes multiple years when new tenant rents fell by more than 3%, as well as multiple years when rents increased by more than 10%. As a result, we avoid having to use only periods of moderate inflation to understand dynamics in high- or low-inflation periods (Nakamura et al., 2018; Montag and Villar, 2023; Gagliardone et al., 2025).

Second, the unit-level detail in our administrative dataset allows us to control carefully for housing quality when analyzing price setting and rental inflation. Because we observe repeated contracts for the exact same unit at different points in the business cycle, we can isolate within-unit price changes and ensure that our results are not driven by shifts in the types of units that transact over time. Because average housing quality has risen and substitution across units can mechanically shift expenditures toward different parts of the quality distribution, failing to hold unit quality fixed can bias inferences about inflation dynamics and pricing frictions

(Molloy, 2024; Orchard, 2025).

Finally, the institutional context of our setting closely resembles a Calvo (1983)-style pricing friction. In Berkeley, rents are tightly regulated once a tenant occupies a unit, and tenant protections are strong. However, landlords are free to set rents without restriction when a tenant vacates.¹ While Calvo pricing is often used as a modeling device for price stickiness, few real-world settings so closely mirror its assumptions. Our data allow us to observe how landlords set prices under these frictions, and our estimates offer novel empirical moments for calibrating Calvo models. Moreover, because 1) landlords face strong restrictions in how much they can adjust rents once a tenant is in place, 2) tenancies often last many years, and 3) high rent growth is typical in the Berkeley market, landlords have strong incentives to set prices optimally when new tenants enter a unit.

Although much of our analysis uses granular administrative data from Berkeley, California, we reproduce the key empirical facts in a nationally representative panel of U.S. rental units for new tenants.² This alignment indicates that the mechanisms we identify extend well beyond the San Francisco Bay Area. At the same time, because Berkeley has experienced greater cyclical and seasonal variation than most U.S. rental markets, it provides a rich setting to study how aggregate shocks influence rent-setting behavior. We use these novel data to document five new facts about the dynamics of rental housing prices.

Our first two facts characterize how landlords use opportunities to reset rents and how their pricing decisions respond to economic conditions. **First**, we provide evidence of nominal rent rigidity for new tenants. The distribution of rent changes shows a pronounced spike at exactly \$0: 13% of tenant spells begin with the same rent as the previous initial contract for that unit. These patterns surprisingly arise in a setting where (1) average rent growth is strong, (2) a new landlord-tenant match eliminates relational contracting concerns, (3) the landlord is already incurring the menu costs of vacancy posting, and (4) landlords have strong incentives to price optimally, due to rent control for continuing tenants. We also find evidence of downward nominal rent rigidity: there is a large mass of rent changes just above \$0 and a missing mass just below.

However, we also find evidence that the strength of these nominal rigidities responds to both economic incentives and the business cycle. The likelihood of a zero rent change strongly correlates with the time between leases, a proxy for the difference between the current price and the optimal reset price. About 25% of rents that reset within one year remain unchanged, compared with 5% after a five-year gap. Thus, landlords are more likely to reoptimize when

¹Throughout our analysis of the Berkeley dataset, we only study (unrestricted) rents for new tenants.

²We proxy for new tenancies by only including observations where the tenants have changed between American Housing Survey waves.

the economic incentives to do so are stronger. We also find that nominal rigidity is asymmetric over the cycle: downward nominal rent rigidity is present during expansions, but not recessions. Downward nominal *wage* rigidity is understood to be important for business cycles, as an unwillingness to cut wages can lead to unemployment. In contrast, since downward nominal rent rigidity is only present in expansions, our results suggest that rent rigidity does not prevent the efficient reallocation of rental units over the business cycle. Thus, while tenant turnover provides opportunities for landlords to reoptimize rents, they do not always do so: the decision to reoptimize depends on the state of the rental market and the gains from adjustment.

Second, we document how landlords adjust rents over the business cycle. We find that state dependence extends beyond the decision to reoptimize: market conditions also shape the direction and magnitude of rent adjustments. Nakamura and Steinsson (2008) show that for the non-shelter components of the CPI, only the frequency of price increases rises with inflation. In contrast, the frequency of price decreases and the size of both price increases and decreases remains stable—patterns consistent with a benchmark menu-cost model. We find strikingly different behavior for the shelter component of CPI. We find that the frequency *and* size of rent increases and decreases respond strongly to the business cycle. We estimate notably similar sensitivities of the frequency and size of price changes to the rental price index in the broader sample of U.S. rental units.³

Together, these two facts provide evidence in favor of models that combine elements of both time- and state-dependent pricing: while exogenous and acyclical tenant turnover provides opportunities for landlords to update their prices, economic conditions determine whether landlords reoptimize and how they adjust rents when they do. These patterns are difficult to reconcile with a pure Calvo (1983) pricing model.

Our first two facts show that landlords' pricing decisions respond strongly to cyclical market conditions. We next ask whether landlords respond similarly to predictable fluctuations in rental demand due to the seasonal cycle. **Third**, we show that predictable seasonal variation in rental demand has large price effects. In both Berkeley and the broader U.S. market, a large share of new leases begin during the summer months. We find that these predictable, seasonal effects have substantial effects on rental prices. However, unlike the response to the business cycle, most of the variation in pricing over the seasonal cycle is driven by differences on the extensive margin: new leases that start in these peak periods are less likely to feature rent cuts and are more likely to include price increases. Rental market seasonality is comparable, perhaps surprisingly, to that found in the home-purchase market, despite the much shorter duration of rental contracts.⁴ Thus, the dynamics of rental price setting depend not just on the

³Our baseline regressions in both the Berkeley and broader U.S. sample include unit fixed effects, and thus our results are not driven by shifts in the quality of units transacted over the business cycle.

⁴Ngai and Tenreyro (2014) document similarly strong seasonality in the home-purchase market and argue

magnitude, but also on the source of the shocks.

Our final two facts examine whether differences across landlords affect both how rents are set and how they respond to changes in market conditions. **Fourth**, we document heterogeneity in how landlords set rents by studying two striking features of the rental-price distribution: round-number bunching and left-digit pricing. In Berkeley, 35% of new contract rents are exact multiples of \$100, and nearly 60% are multiples of \$50. We find similarly large bunching in the broader U.S. market, indicating that price clustering is a widespread feature of rental markets, and not unique to Berkeley, CA.⁵⁶ We also document pronounced left-digit pricing, with 15% of rents set at prices exactly \$5 below multiples of \$100. We find that both rounding and left-digit pricing vary systematically with landlord characteristics: large, corporate landlords rely less on round-number bunching but more frequently employ left-digit pricing. We confirm that these relationships are causal by using variation induced by sales of rental properties to landlords of different sizes. We find that after a property is sold to larger landlords, the amount of coarse pricing falls and the amount of left-digit pricing increases, providing evidence that these relationships are causal, and are not driven by other types of unobserved heterogeneity. We interpret this pattern as evidence that smaller landlords, who exhibit more coarse pricing, are more prone to behavioral “pricing mistakes.” The persistence of such mistakes suggests that market power and search frictions permit substantial heterogeneity in pricing behavior. Overall, these results indicate that behavioral frictions on the seller side play a meaningful role in rental price-setting, even in a high-stakes environment where prices may remain fixed for many years.

We also find strong evidence that these behavioral pricing patterns vary with the complexity of the pricing problem. When rent inflation is unusually high or low, landlords shift toward round-number pricing (multiples of \$50/\$100) and away from left-digit pricing. In addition, when landlords reset prices after long tenancies—and therefore face larger optimal price adjustments—they rely more heavily on coarse pricing and less on left-digit pricing. These effects are especially pronounced among small landlords, though they do affect large landlords as well. These results suggest that unusual nominal adjustments raise cognitive demands and reduce pricing precision. They also point to a novel cost of inflation volatility that is omitted

that it is inconsistent with standard housing models, suggesting instead that thick-market effects that raise match quality amplify seasonal moving patterns. Finding comparable (or even larger) seasonality in the rental market, despite shorter contract durations and weaker incentives to maximize match quality, suggests that other forces are necessary to explain seasonality across both rental and owner-occupied markets.

⁵These magnitudes substantially exceed prior U.S. estimates from Genesove (2003), which relied on survey data from the 1970s and 1980s.

⁶Our results thus parallel recent findings from Dube, Manning and Naidu (2025), who find comparable bunching in the wage distribution and show it reflects a combination of monopsony power and employer misoptimization.

from standard New Keynesian models: as inflation becomes more volatile, the complexity of price setting increases, leading to less precise pricing.

Fifth, we show that landlord heterogeneity is also crucial for understanding rent dynamics over the business cycle. We find that large landlords adjust prices more aggressively in response to cyclical variation: they are more likely to cut rents during recessions and raise rents, and by larger amounts, during expansions. We find very similar results when we use property sales as a source of quasi-experimental variation in landlord size. Our results suggest that firm heterogeneity is a critical driver of changes in inflation over time, implying that as more large landlords enter the rental housing market, rental price inflation will become more volatile and cyclical. Together, our final two facts show that landlord identity shapes both the precision of rent setting and the responsiveness of rents to aggregate conditions.

We use these new facts to discipline a model of rental price setting. Our model features heterogeneous landlords that set prices under imperfect information and face costs of actively reoptimizing and cutting rents. We use the estimated model to study how the entry of large landlords into the rental housing market affects shelter inflation for new tenants. We find that replacing current small landlords with large landlords would increase new-tenant rents on average by 0.40 p.p. per year while also substantially increasing the volatility of rental prices. In particular, we estimate that in boom years, rental price inflation would be about 1 p.p. higher than in the status quo, and this would be slightly offset by lower rent growth in bust years. Overall, our results suggest substantial effects of changes in the composition of landlords on the level and volatility of shelter inflation.

Taken together, our facts provide new evidence that can discipline static and dynamic models of housing markets. Our results also speak to a wide range of crucial policy issues, including the role of housing in the evolution of inflation dynamics over the business cycle and cost-of-living pressures over time. Our work also has implications for recent policy and academic debates about the welfare impacts of the growing influence of larger, corporate landlords.

1.1 Literature

Our work contributes to three main strands of the literature. First, we build on foundational research documenting price-setting behavior for non-housing components of the CPI. Studies such as Hosken and Reiffen (2004), Bils and Klenow (2004), Nakamura and Steinsson (2008), Klenow and Kryvtsov (2008), and Klenow and Malin (2010) establish empirical facts that have shaped how economists model pricing frictions. In contrast, we present new statistics on shelter prices—the largest component of household expenditure and a key driver of recent inflation dynamics. Our institutional setting closely resembles the staggered pricing friction introduced

by Calvo (1983). While Calvo pricing is often invoked for tractability, few real-world environments closely match its structure. Bils and Klenow (2004) and Nakamura and Steinsson (2008) show that many observed facts are more consistent with menu-cost models. In contrast, we offer new moments that describe firm behavior under Calvo-style timing frictions. Our results also point to inattentiveness or cognitive costs of reoptimizing that lead to patterns that deviate from these standard time-dependent models. We also document state-dependent pricing in the rental market, contributing to recent work on how price-setting behavior evolves in high- and low-inflation environments (Golosov and Lucas Jr., 2007; Nakamura et al., 2018; Montag and Villar, 2023; Gagliardone et al., 2025). We document the effects of seasonality in the rental market, contributing to a broader literature that leverages seasonal patterns to study economic behavior (Barsky and Miron, 1989). Our work on how rents vary over the business cycle can also inform recent papers examining the optimal policy response to housing price inflation (Bianchi, McKay and Mehrotra, 2024).

Second, we contribute to the literature on rental price dynamics. Prior work has documented nominal rent stickiness using household surveys (Genesove, 2003; Hoffmann and Kurz-Kim, 2006; Shimizu, Nishimura and Watanabe, 2010; Aysoy, Aysoy and Tumen, 2014; Gallin and Verbrugge, 2019), but such data often suffer from recall error and limited coverage. In contrast, we leverage an administrative dataset based on legally reported contract rents submitted by landlords. We also study rent setting over decades, which allows us to analyze how rental-price setting is affected by cyclical variation. We provide evidence of how contract rental prices respond to the business and seasonal cycle. We also show that nominal rigidity and coarse pricing are endogenous responses to inflation, the timing of lease resets, and landlord characteristics. Our results highlight behavioral frictions as key drivers of sticky rent-setting behavior.

While our analysis focuses on rents for new tenants, we complement recent work by Adams et al. (2024), who emphasizes the differences between pricing for new and continuing tenants.⁷ Calder-Wang and Kim (2024) study how the adoption of AI pricing tools affects asking rents using a large sample of multi-unit buildings. Our work also relates to Genesove and Mayer (2001), who document how behavioral frictions influence home sale and purchase decisions. Baker (2025) uses the same Berkeley dataset as we do to study pass-through of property taxes to rents, and finds results consistent with non-standard, behavioral pricing patterns.⁸ Finally, our work complements a significant recent literature on the effects of large, institutional landlords (Lambie-Hanson, Li and Slonkosky, 2022; Gurun et al., 2022; Neroli, 2022; Hanson,

⁷Ball and Koh (2025) also study the discrepancy between new-tenant rents and rents for continuing tenants.

⁸As in Mondragon et al. (2026) and Baker (2025), our work also contributes to work on the Bay Area housing market.

2024; Barbieri and Dobbels, 2025; Chang, 2025; Coven, 2025; Gorback, Qian and Zhu, 2025; Raymond, 2025).

Finally, our findings relate to a growing literature on behavioral frictions in firm decision-making (Matějka, 2016; DellaVigna and Gentzkow, 2019; Stevens, 2020; Strulov-Shlain, 2022, 2024). We show that behavioral considerations, including heuristic pricing and inattention, are first-order in the rental housing market, and that their influence varies systematically with the identity of the seller and the business cycle. Repetto and Solís (2019) document left-digit pricing in the Swedish home purchase market. Yang (2022) shows that firms with greater product scope are more informed about inflation, while Bhattarai and Schoenle (2014) and Yang (2022) show that multiproduct firms adjust prices more regularly. Amiti, Itskhoki and Konings (2019) provide evidence that the degree of strategic complementarities varies across firms. In contrast to this focus on strategic complementarities, we provide evidence consistent with larger landlords having better information and being more responsive, conditional on a price reset, to business cycle conditions. In related, contemporaneous work, Park (2024) finds heterogeneity in how frequently landlords update listed rents for vacant units. The paper is also closely related to contemporaneous work by Gallin et al. (2025), though that paper focuses more on differences in rent-setting behavior for new and continuing tenants. In contrast, we study rent-setting for new tenants, which more closely maps to a Calvo setting by eliminating relational contracting concerns. An advantage of our setting relative to this work is also that we focus on heterogeneity across landlords, and how that shapes price-setting in the rental housing market. Consistent with Dube, Manning and Naidu (2025), who interpret round-number bunching in wages as a joint outcome of firm misoptimization and market power, we document pronounced bunching in the rent distribution and interpret it as reflecting a combination of landlord misoptimization and pricing power.

2 Setting & Data

2.1 Berkeley Rent Registry

We use administrative data on contract rents in Berkeley, California from the Berkeley Rent Board covering the period from 1980–2022. The Rent Board collects detailed information on rental units subject to rent control or eviction protections, including unit characteristics, initial rent, and lease start dates. Landlords in Berkeley who are partially or fully covered by rent control are required to register any new leases with the Rent Board. Our primary sample focuses on rents for new tenants from 1999 onward, after the adoption of vacancy decontrol under the Costa-Hawkins Rental Housing Act of 1995. Under vacancy decontrol, these new-

tenant rents are freely set by landlords and are not subject to rent control restrictions. Our sample excludes many single-family homes and condos which are not subject to rent control, and thus were not required to report starting rents until very recent years.

We exclude the small number of units with more than five bedrooms, which tend to either be very large homes for students (e.g., fraternity and sorority houses) or are the wealthiest segment of the rental market. The dataset covers 19,000–26,000 units annually, with the number increasing as more units were built and/or registered with the Rent Board.

We link Rent Board records to property tax data from the Alameda County Assessor’s Office, which provides owner names and mailing addresses. Corporate landlords are identified by terms like LLC, Corp, or Company in the owner name (Harwood, Gould Ellen and O’Regan, 2025). We group properties by owner mailing address to construct a proxy for landlord size. We also use publicly available permit data from the City of Berkeley, linking each permit to its corresponding rental unit and lease spell (when appropriate). We also utilize permit score data from Baker (2025), in which natural language processing methods score each permit for its rent-increasing potential.

Table 1, Panel (A) presents summary statistics from the Berkeley sample. The vast majority of units in our data are relatively small units (studios to two bedroom units). The median observation in our sample is for a lease for a one-bedroom apartment in a twelve-unit building, owned by a landlord who owns 28 units in Berkeley. The median tenant spell in our sample lasts two years, but with significant dispersion: the 90th percentile time between new leases in our sample is almost 6 years. This highlights the strong incentives landlords in our sample have to set prices correctly: on average, a landlord will not be legally allowed to fully update their price for almost 3 years.

2.2 Berkeley Rental Market

Berkeley, California, is part of the San Francisco Bay Area, one of the most competitive rental markets in the U.S., with persistently low vacancy rates around 5%.⁹ Consistent with this, rents in our Berkeley data are quite high. The median nominal rent in our main sample period is \$1,495 per month. Figure A.1 plots an annual repeat-rent index for new-tenant rents in Berkeley since 1999. The index includes unit fixed effects, capturing within-unit price changes over time and adjusting for time-invariant quality. Quality-adjusted rents rose by around 130% during our sample period, though growth was uneven. New-tenant rents declined after the 2001 and 2008 recessions and were flat from 2018 to 2021.

Rent-setting in the Berkeley market closely mirrors the pricing friction introduced by Calvo

⁹Housing Vacancies and Homeownership (CPS/HVS), U.S. Census Bureau.

(1983), combined with partial indexation. Landlords can freely set rents when a tenant exits, but face strong limits in how much they can raise rents once a tenant is in place.¹⁰ In particular, each year the rent board establishes an Annual General Adjustment, which is equal to the maximum of either 65% of the Bay Area CPI increase or 5%. In addition, AGA adjustments are not allowed in the year a tenant enters a rental unit, or the calendar year after that. Strong tenant protections in Berkeley further limit evictions and, in some cases, restrict rent increases following an eviction. Because landlords reset prices precisely when they must already re-list the unit and match with a new tenant, frictions such as menu costs or buyer-seller relationships are minimal. This makes our setting a clean environment in which to study price-setting behavior under Calvo frictions. Throughout our analysis, we focus on rents for new tenants, and on changes in initial rents paid by tenants.

2.3 American Housing Survey

We use the American Housing Survey (AHS) to test whether our findings generalize to the broader American rental market. The AHS is a longitudinal panel survey conducted by the Census Bureau that tracks individual housing units over time.¹¹ We focus on the 1985 panel, which follows the same houses every other year from 1985 to 2013.¹²

Despite its broader scope, the AHS has several critical limitations that led us to use the Berkeley registry as our primary data source. First, the AHS does not capture all new contracts, even for sampled units, because the survey is conducted biennially. Second, the AHS does not have a separate identifier that can be used to differentiate between new tenant contracts and continuing tenant contracts, a distinction crucial for analysis as these contract types may differ systematically (Adams et al., 2024).¹³ Third, the AHS provides less business cycle variation in rents because the aggregate U.S. rental market is less volatile than the San Francisco Bay Area, observations occur only biennially, and the AHS lacks detailed geographic information to precisely locate properties. Fourth, the AHS contains no data on landlord characteristics, which our Berkeley analysis reveals as a significant driver of pricing behavior. Finally, as a survey completed by tenants, the AHS may suffer from non-response bias and imprecise recall of rental contract details.

To construct our main sample, we use AHS data from 1985 to 2013, focusing exclusively on rental properties and excluding mobile homes to enhance comparability with the Berkeley sam-

¹⁰We also show in Section 3.2 that the number of new tenant contracts does not vary with the business cycle.

¹¹New homes are added to the sample as they are constructed.

¹²We do not use data from the 2015 panel onwards because many crucial variables are either suppressed for confidentiality or rounded, making it impossible to study questions related to coarse pricing, left-digit pricing, or sticky rents.

¹³However, we impute these changes using differences in the households sampled across survey waves

ple. We restrict our analysis to metropolitan areas, including center cities and urban suburbs, while excluding households in rural areas. We link units longitudinally to measure changes in rents over time. To better approximate our new tenant sample, we only include observations where the current household differs from the one interviewed in the previous survey wave. All observations are weighted using the Census Bureau’s sampling weights.

Table 1, Panel (B) presents summary statistics from the AHS sample. Given the earlier time period (beginning in the 1980s), and the cooler market in the rest of the United States, the median rental price in this sample is much lower, at around \$550. The median unit in the broader sample is a 2-bedroom unit, in a building with 6 units.

3 How Often and How Much Do Rents Change

3.1 Fact 1: Downward Nominal Rent Rigidity

Our first fact is that rents exhibit pronounced nominal rigidity, even when tenant turnover provides landlords with an opportunity to reoptimize rents without any restrictions. Figure 2, Panel (A) plots the distribution of rent changes for new tenants at each nominal dollar value.¹⁴ A large spike appears at exactly zero: in our main sample, 13% of new-tenant leases have rents that are identical to the initial rent paid by the previous tenant. This is the same share reported by Genesove (2003), despite their study using a different dataset and a sample period two decades earlier.

Figure 2, Panel (A) also provides initial evidence of downward nominal rent rigidity. The histogram of rent changes features a striking asymmetry: there are many more rent changes right above zero than right below. For example, there are nearly 3.5 times more rent changes of exactly \$10 than there are rent changes that are -\$10. Panel (B) plots the distribution of rent changes after removing new rental contracts where initial rents are unchanged. Panel (B) shows evidence that rents display the hallmarks of downward rigidity: rents increase much more often than they fall and “pile up” close to zero, as evidenced by the fact that there are far more very small rent increases than there are decreases. These patterns closely resemble those documented in wage-setting, where small increases are common and small decreases are rare (Hazell and Taska, 2025). In the labor market, this asymmetry is often attributed to workers’ aversion to wage cuts (Davis and Krolkowski, 2025). However, in the rental market, that mechanism is absent (it is implausible that tenants would object to rents falling), suggesting that downward nominal rigidity may also arise from firm-side frictions, such as cognitive constraints, inattention, or firms’ own desire not to cut prices.

¹⁴Rent changes are rounded to the nearest dollar.

Figure 2, Panel (C) shows that this rigidity responds to economic incentives. We exploit the fact that the cost of not updating rents rises with the time between leases, as market prices drift away from the previously set rent.¹⁵ Figure 2, Panel (C) shows, correspondingly, that the probability of the rent remaining fixed after a new tenant enters declines sharply with the time between leases. This relationship is estimated after residualizing on neighborhood-by-year and unit fixed effects, thereby controlling for unit-specific factors, the business cycle, and neighborhood trends. This pattern suggests that landlords face a cost of reoptimizing and are more willing to incur it as the potential pricing error grows. Because the landlord must pay the fixed cost of finding a new tenant regardless of whether or not they change the rent, this reoptimization cost is distinct from the standard cost of posting a new vacancy. We estimate a semi-elasticity of -0.11, implying that a 50% increase in the length between leases lowers the probability of the landlord leaving the rent unchanged by about 4.5 percentage points.

We also find that rigidity responds to economic incentives that arise due to the business cycle. Figure 2, Panel (D) plots the kernel density of rent changes (excluding zeros) separately for different phases of the business cycle, as represented by the change in the new-tenant rental index. The asymmetry near zero—a concentration of mass just above zero and missing mass just below—is primarily present during expansions. In contractions, this asymmetry decreases significantly. This suggests that landlords’ reluctance to decrease rental prices is primarily present during periods of high and moderate rental price growth. During periods of low rent price growth, landlords are willing to set lower rents; indeed, there is very little asymmetry or missing mass in the rent change distribution for the lowest rent growth tercile. This form of state-dependent rigidity contrasts with evidence of downward nominal wage rigidity, where constraints appear binding and firms do not cut wages even in downturns (Hazell and Taska, 2025).

Appendix Figure C.1 reproduces our core findings using the nationally representative AHS. Panel (A) plots nominal rent changes (rounded to the dollar) and reveals a pronounced spike at \$0, consistent with the Berkeley evidence. In the AHS, roughly 8% of rents are unchanged, versus 13% in Berkeley.¹⁶ The smaller zero mass likely reflects measurement frequency: the AHS observes rents biennially, which means we do not observe many rent changes that occur after the shortest leases, which we show contributes disproportionately to bunching at \$0.¹⁷

Figure C.1, Panel (B) reports the kernel density of nominal rent changes excluding zeros.

¹⁵Appendix Figure A.2 illustrates this graphically by showing a binscatter plot of the relationship between the months since the previous lease began and the reset price. There is a tightly estimated positive slope, with an estimated elasticity of the reset price to the time between leases of 0.14.

¹⁶All AHS estimates restrict to units that experience household turnover between waves; zeros are thus not attributable to continuing tenants.

¹⁷See Figure 2, Panel (C).

Mirroring the Berkeley results, Panels (A) and (B) display signatures of downward nominal rigidity: missing mass just below zero and excess mass just above. Panel (C) documents the same state dependence found in the Berkeley rental market. Rent rigidity is strong in periods of rapid national rent growth and is weaker in moderate or low-growth periods. The patterns in the national rental market are attenuated relative to Berkeley, possibly due to the fact that the broader U.S. rental market is less volatile than the market in the Bay Area. Overall, the results from the AHS corroborate that our findings extend beyond the Bay Area, while also underscoring the value of the granular Berkeley data and the elevated volatility of the primary market we study as useful identifying variation for studying landlords' pricing behavior over the business cycle.

One potential concern with our results is that they are distorted by the fact that Berkeley does allow small rent adjustments. To address this concern, in Appendix Figure A.4 we reproduce our main facts using a sample of leases that were not eligible for any rent adjustments under Berkeley rent control regulation because the lease ended before any adjustments were allowed. We find the exact same qualitative patterns documented in Figure 2. In this smaller sample the number of \$0 rent changes is higher (22% vs. 13%), though this would naturally be expected given that these leases were shorter, and thus gaps between incumbent and optimal reset prices should be smaller. In addition, the estimated relationship between rent rigidity and the time between leases shown in Panel (C) is actually **stronger** in this sample than in the overall sample, suggesting that the relationship is not a mechanical effect of partial indexation. The similarity of the results across these two samples, plus the fact that these results are also found in the broader U.S. sample, suggest that these patterns are not mechanically driven by rent indexation.

3.2 Fact 2: How Rents Adjust to the Business Cycle

Section 3.1 examined how the likelihood that landlords actively reoptimize new-tenant rents depends on the state of the market; we next examine more broadly how landlords' rent-setting behavior varies over the business cycle. Nakamura and Steinsson (2008) document that the frequency of price increases comoves with inflation for the non-housing components of the CPI, while the magnitude of increases, and both the frequency and magnitude of decreases, is mostly acyclical. These patterns align with the predictions of canonical menu-cost models.

In contrast, the housing component of inflation displays strikingly different patterns. Despite stability in total leasing volume over the business cycle,¹⁸ the composition of rent changes

¹⁸Appendix Figure A.3 shows that the number of new tenant leases in our sample is acyclical. Lease counts are flat from 2002 to 2013, decline modestly thereafter, and fall sharply during the COVID-19 pandemic before rebounding in 2021. This acyclicity implies that tenant mobility—and thus opportunities for rent resetting—is

is highly cyclical. Figure 3 depicts the evolution of the size and frequency of rent adjustments among units turning over within two years. Panel (A) reveals a strong correlation between the business cycle and the share of rent increases, decreases, and unchanged rents. During periods of weak rental growth, rent decreases outnumber increases for units that turn over within two years, and over 20% of these contracts show unchanged rents.¹⁹ In contrast, in expansionary periods, more than 80% of new leases record rent increases, and rent cuts are virtually nonexistent, consistent with downward nominal rent rigidity.

Table 2, Panel (A) formalizes these patterns. Column (1) confirms that the aggregate number of new leases is uncorrelated with the rent index. Columns (2)–(4) document that the number of rent increases, decreases, and unchanged rents all exhibit strong systematic correlation with rent price inflation. All estimates are large, highly significant, and feature high R^2 values, implying that cyclical conditions explain a substantial share of variation in the composition of rent changes over time.

Figure 3, Panel (B) demonstrates that the intensive margin of rent adjustments—the size of price changes—also exhibits pronounced cyclical variation. As with the frequency of price changes, the size of both rent increases and decreases covaries with rental price inflation in the expected directions—the size of rent increases (decreases) covaries positively (negatively) with inflation. In the tightest markets, such as before the tech bust of 2001 and during the mid-2010s expansion, the median rent increase exceeds 10 log points. In contrast, in the weakest markets, the median rent increase falls to around 5 log points. The median rent decrease also varies over the business cycle, though its time-series variation is much less pronounced than that of rent increases.

Table 2, Panel (B) confirms these patterns using time-series regressions. Columns (1) and (2) report estimates of the effect of the rent index on the median size of rent increases and decreases for leases that turn over within two years. We find statistically significant correlations between the rent index and the size of both rent increases and decreases, with the effect being more than two times larger for rent increases. Columns (3) and (4) show nearly identical coefficients in a sample including all new leases. Overall, we find strong evidence that both the frequency and magnitude of rent increases and decreases covary with inflation, contrasting sharply with the findings of Nakamura and Steinsson (2008) for non-housing CPI components. However, rent decreases covary with inflation substantially less than rent increases.

Table 3, Panel (A) presents the results for the individual, lease-level data in the Berkeley sample, after residualizing on unit-level fixed effects. We use unit-level fixed effects to identify responses by comparing rent changes for the same unit when it turns over at different points

primarily governed by idiosyncratic household shocks rather than aggregate conditions.

¹⁹Note that periods of weak rental growth are delayed relative to recession starts.

in the business cycle. This specification isolates variation arising from macroeconomic conditions rather than from differences in unit quality or composition. This also ensures that our results are not driven by changes in relative demand across market segments over the cycle. Columns (1) – (3) examine the extensive margin. Consistent with the aggregate analysis, the likelihood of a rent increase rises with the rent index, while the likelihood of a rent decrease or an unchanged rent declines.²⁰ Columns (4) and (5) show the effects on the magnitude of price changes (in log terms). Column (4) shows that, when including unit fixed effects, the elasticity of rent increases with respect to the rent index is 0.7, compared with 0.5-0.6 in the aggregate data. Column (5) shows that the elasticity of rent decreases with respect to the rent index is equal to -0.3 (versus -0.2 in the aggregate data). Overall, both the extensive and intensive margins of price changes respond strongly to the business cycle, even after controlling for time-invariant unit quality.

Table 3, Panel (B) replicates this analysis for the broader U.S. sample. Once again, we use detailed unit fixed effects to study how pricing behavior responds to the business cycle, while carefully controlling for compositional changes and the quality of transacted units. Overall, we estimate elasticities that are closely aligned with those from Berkeley. Columns (1) - (3) show the effects on the extensive margin, which are nearly identical in magnitude to those found in the Berkeley sample. The effects are highly statistically significant, though standard errors are about 2-3x larger—likely due to both higher sampling variance in survey data and heterogeneity in local markets not reflected in our national price index. Columns (4) and (5) show effects on the intensive margin: the elasticity of rent increases is about 0.9, compared to 0.7 in the Berkeley sample, while the elasticity for rent decreases is imprecisely estimated with large standard errors. Taken together, the unit-level results confirm that rental inflation is driven by adjustments on both the extensive and intensive margins in both samples, even after carefully controlling for unit quality.

3.3 Discussion of Facts 1 and 2

A natural question is what types of models of price-setting are consistent with our first two empirical facts. Our setting closely matches a traditional Calvo environment: turnover generates exogenous, acyclical opportunities to change prices that are not driven by the business cycle. We also find a strong positive relationship between the size of rent changes and the time between new leases, which is consistent with time-dependent pricing. In addition, the size and frequency of both price increases and decreases covary with inflation. This contrasts

²⁰Because the dependent variables are binary indicators for specific types of price changes, coefficient magnitudes are not directly comparable to those in Table 2. Instead, these estimates are semi-elasticities of indicators for different extensive-margin rent adjustments to the business cycle.

with the findings from Nakamura and Steinsson (2008), who find that for the non-housing CPI, inflation mainly covaries with the frequency of price adjustments rather than their size, in line with menu-cost models. Taken together, these facts are consistent with time-dependent pricing playing an important role in rent setting.

At the same time, several features of our data are hard to reconcile with a pure Calvo model. First, we observe a pronounced spike at exactly zero in the rent-change distribution at turnover, whereas standard time-dependent models with full reoptimization imply essentially no mass at zero when a reset opportunity arrives. Second, the probability of leaving rents unchanged at turnover is state-dependent: zeros rise in weak rent markets, shrink in strong rent markets, and fall sharply with the time between leases (i.e., with the size of the price gap), pointing to a cost of reoptimizing that landlords pay when the benefit is large. Thus, our first two facts are naturally consistent with hybrid models, which combine an exogenous reset opportunity with a fixed cost to reoptimize in that period.²¹ Importantly, this cost is distinct from traditional menu/vacancy costs: a vacancy must be posted either way, so the residual margin more likely reflects information or managerial (cognitive) costs. Given that the median lease duration is about two years, failure to reoptimize can be costly, and is consistent with sizable information or cognitive costs.

A final question is whether these time-dependent features are generalizable beyond the Berkeley context, where rent control for continuing tenants is crucial for matching the Calvo institutional setting. However, the fact that we replicate key empirical patterns—the strong correlation of both the frequency and size of rent increases with inflation, the pronounced rent-change spike at zero, and the state-dependence of the spike at zero—in a nationally representative survey suggests that our findings provide useful guidance for models of the broader U.S. housing market. In addition, the fact that rents for continuing tenants have been found to be sticky even outside of strict rent control implies that, more generally, landlords rarely reoptimize prices without a new lease, reinforcing the importance of time-dependent mechanisms in rental price setting (Adams et al., 2024; Ball and Koh, 2025).

3.4 Time-Varying Unit Quality

Our baseline regressions using unit-level data include unit fixed effects, which removes persistent differences in unit quality. However, these controls cannot account for time-varying changes in unit quality, driven by endogenous renovations or maintenance. For example, if landlords are less likely to improve units in downturns, our cyclical results could be biased

²¹This contrasts with canonical “Calvo-Plus” models which combine exogenous reset timing with a menu cost that firms can pay at any time to reoptimize prices (Nakamura and Steinsson, 2010; Alvarez, Lippi and Oskolkov, 2021).

as a result of not properly accounting for changes in unit quality.

To address concerns that renovations may vary with the business cycle and thus impact our results, we use data on building permits to test for how renovations and maintenance vary over time. Figure A.7 plots total building permits in Berkeley. Permit activity appears highly seasonal, with activity increasing in the summer months. However, a cyclical relationship is less obvious: permit activity appears stable from 1995 to 2004, decreases from 2004 to 2010, increases throughout the 2010s, and falls sharply during the Covid-19 pandemic. We directly test for the relationship between permit activity and the Berkeley rent price index in Table B.1. Column (1) shows that the total number of permits is uncorrelated with the rent price index, and Column (2) provides a similar insignificant result for the log number of permits. This result provides further evidence that permit activity does not drive the cyclical behavior of rent prices.

As an additional test, we create a permit-adjusted repeat rent index, using the permit-scoring method from Baker (2025). Baker (2025) uses permit data from the City of Berkeley and matches each permit to its corresponding rental unit. Natural language processing methods are then used to score each permit for its rent-increasing potential. We find almost no difference between the non-permit-adjusted and permit-adjusted repeat rent index, shown in Figure A.6. The similarity of the two indices provides reassurance that cyclical, time-variation in renovations do not drive our results.

3.5 Fact 3: How Rents Respond to the Seasonal Cycle

Our first two facts examined how landlords adjust rents in response to changes in rental demand over the business cycle. The seasonal cycle provides a second, predictable source of variation in rental demand that we use to study price-setting behavior. For our third fact, we document how seasonality shapes housing demand, pricing, and rent changes. Appendix Figure A.5 displays the number of lease starts by month over the sample period in both Berkeley and the broader U.S. Two patterns emerge. First, in Berkeley, most leases are signed during the summer months between June and August. Second, lease activity drops sharply between October and April. While Berkeley's status as a university town likely contributes to the summer spike in demand, this pattern is not unique: nationally, 32% of leases begin between June and August.²²

The timing of lease start dates also affects rent prices. Figure 4 plots monthly rental premia on the average number of leases in each month in both Berkeley and the U.S. sample.

²²Appendix Figure A.5, Panel (B) compares the level of seasonality in the rental market to the degree of seasonality in the U.S. home purchase market documented by Ngai and Tenreiro (2014). We find comparable seasonality in both markets.

The monthly premia are calculated using both neighborhood-by-year (or for the U.S. sample, city-by-year) and unit fixed effects, and thus strip out local trends and time-invariant differences across individual units. In Berkeley, this amounts to comparing rents for the same unit within the same neighborhood and year, while also holding fixed the time between leases and the month in which the previous lease began. We estimate positive and significant slopes in both samples. Thus, months where the market is ‘hotter’ (i.e., months where there are many new leases, such as the summer months) command higher initial rents than cooler months. Quantitatively, the hottest months in the broader U.S. market are, on average, associated with rents about 3% higher than the coolest months, while in Berkeley the difference between the coolest and hottest months is around 7%.

We next study how landlords adjust prices in response to seasonal variation. Table 4 decomposes adjustment at turnover into extensive and intensive margins.²³ In contrast to the dynamics over the business cycle, most seasonal adjustment in both Berkeley and the U.S. occurs through changes on the extensive margin (i.e., whether a unit experiences rent increases versus rent decreases), rather than on the intensive margin (i.e., the size of the rent change, conditional on its sign). Table 4, Panel (A) shows that, in peak months, landlords are more likely to raise rents and less likely to decrease rents. Panel (B) shows similar patterns in the broader U.S. sample: in peak months, the share of rent increases rises, while the shares of decreases and unchanged rents fall. The estimated elasticities are somewhat smaller in Berkeley than in the national data.

Columns (4) – (5) in both panels of Table 4 report intensive-margin responses. In both samples we estimate null effects of the seasonal cycle on the size of rent increases, conditional on increasing. In Berkeley, we find larger rent cuts in weaker months, conditional on decreasing; in the U.S. sample, the corresponding estimates are imprecise and not statistically different from zero. Aggregating, we estimate elasticities of new-tenant rents with respect to market tightness (proxied by the number of lease starts) of about 0.02 in Berkeley and 0.04 in the U.S.

3.6 Discussion of Pricing Seasonality

Overall, we find substantial effects of the seasonal cycle on rental prices. Our seasonality estimates mirror Ngai and Tenreyro (2014) for home purchases. Ngai and Tenreyro (2014) argue that the large predictable variability in home prices is inconsistent with standard models and requires idiosyncratic preferences and thick-market effects. Because rental contracts are far less durable and thus confer much more modest match-specific surplus, the pronounced seasonal rent premia are unlikely to arise from match effects alone. Our results suggest that

²³These regressions employ the same fixed effects as those shown in Table 3.

additional mechanisms must operate alongside any match effects to generate the observed patterns in both rental and home-purchase markets.

We also find differences between seasonal and business-cycle responses. Over the business cycle, landlords respond on both the extensive and intensive margins. In contrast, most of the seasonal variation in rent prices is driven by the extensive margin (the likelihood of increases versus decreases), with limited movement in the size of increases. Thus, our results suggest that how landlords respond to demand shocks depends on both the size and the source of the shocks. Finally, we estimate slightly smaller seasonal elasticities in Berkeley than in the broader U.S. rental market. Because the seasonal cycle is larger in Berkeley than in the aggregate U.S. economy, this suggests that responses may be nonlinear with respect to the size of the shock.

4 Landlord Heterogeneity Drives Inflation Dynamics

We next document substantial heterogeneity in landlords' pricing behavior, both on average and over the business cycle. These patterns highlight the important role of firm heterogeneity in shaping inflation dynamics and show how secular shifts in firm composition can alter aggregate inflation.

4.1 Fact 4: Heterogeneous Landlord Price Setting

Our fourth fact is that landlord heterogeneity is an important predictor of rent setting. We focus on landlord heterogeneity by size, proxied by the number of units an individual landlord owns in Berkeley.²⁴ To characterize how rent-setting behavior differs across landlords, we begin with two salient features of the rental price distribution: round-number bunching and left-digit pricing. Figure 5, Panel (A) plots the distribution of rents by the distance of each rent from the nearest multiple of \$100. Thus, an observation at 0 is a multiple of 100; an observation at -50 is a multiple of 50 (but not 100); an observation at -25 is \$25 below the nearest multiple of 100 (e.g., \$1,175 or \$1,375). We plot this separately for large landlords (owning 75 or more units) and small landlords.

Panel (A) documents substantial bunching in the Berkeley rent distribution. Three patterns stand out. First, a large share of rents bunch exactly at multiples of 100. In our main sample, around 35% of observations fall at these round numbers, which is similar to the share of wages ending in round numbers reported in the CPS (Dube, Manning and Naidu, 2025). Second, there is a smaller but still significant spike at rents that are multiples of 50 but not 100; in

²⁴We also look at heterogeneity by corporate status, proxied by whether the owner's recorded name contains terms such as "LLC" or "Corp.", and find similar results.

our baseline sample, about 24% of observations fall into this category. We also find smaller bunching at finer increments (multiples of 25, 10, and 5). In the full sample, 97% of initial rents are at multiples of 5, and 99% are whole dollars. These bunching rates are substantially higher than those documented in earlier survey data. Genesove (2003) finds that 7–11% of rents were set at multiples of 100, and 8–12% at multiples of 50 (but not 100) in the 1970s and early 1980s. In contrast, our estimates are more than three times higher for multiples of 100 and about twice as high for multiples of 50. Both Genesove (2003) and Aysoy, Aysoy and Tumen (2014) argue that a substantial portion of nominal rent rigidity can be explained by landlords using coarse pricing patterns, where landlords set rents at multiples of round numbers. However, since we observe the same level of rigidity despite finding a much larger degree of price clustering, our results suggest that coarse pricing alone is not sufficient to explain nominal rent stickiness.

Third, the rent distribution shows clear evidence of left-digit pricing (Anderson and Simester, 2003; Lacetera, Pope and Sydnor, 2012; Strulov-Shlain, 2022, 2024). There is a notable spike exactly \$5 below the multiple of \$100, with the mass at this point equal to roughly 42 percent of the spike at the corresponding multiple of \$100. More broadly, we observe excess mass just below round numbers and missing mass just above. There are roughly 39.5 times as many observations in the interval $x \in [-9, 0)$ as in $x \in (0, 9]$, where x is the distance from the nearest multiple of \$100. This suggests that landlords believe left-digit bias significantly influences tenant behavior. To our knowledge, this is the first paper to document left-digit pricing in the rental housing market.

Figure 5, Panel (B) shows that this bunching varies substantially by landlord characteristics. Panel (B) shows that large landlords are much less likely to set rents at multiples of \$100 and are much more likely to set rents just below multiples of \$100. Only about 25% of rents set by large landlords are multiples of \$100, compared to more than 40% of rents set by small landlords. In contrast, almost 30% of rents set by large landlords are set at amounts that are \$5 less than multiples of \$100, compared with only 10% for small landlords.²⁵

Figure 5, Panels (C) and (D) shows that the propensity to bunch varies smoothly with landlord size. Panel (C) plots a binscatter of the share of rents set exactly at multiples of \$100 against log landlord size, residualized on neighborhood-by-year fixed effects.²⁶ The relationship is tightly estimated and negative. We estimate a semi-elasticity of bunching of -0.05 , implying that for the smallest landlords, nearly 50% of rents are set at \$100 multiples, while

²⁵Appendix Figure A.11 shows similar patterns when we split the sample into corporate and individual landlords. Corporate landlords are about 10 p.p. less likely to bunch at multiples of \$100, and about 10 p.p. more likely to set rents exactly \$5 below a multiple of \$100.

²⁶The inclusion of these fixed effects ensures that the pattern is not driven by cross-market composition or business-cycle heterogeneity.

for the largest landlords the degree of bunching is roughly half as large.

Panel (D) reports the mirror pattern for left-digit pricing. The semi-elasticity with respect to size is 0.06, implying that the smallest landlords engage in essentially no left-digit pricing, whereas for the largest landlords nearly 35% of rents are set just below \$100 thresholds.

One potential concern is that these findings reflect some other type of unit-specific unobserved heterogeneity that is correlated with landlord size. As a result, we use sales of rental properties as a source of quasi-experimental variation in landlord size. We subset to a set of properties that are sold at some point during our sample period. We then estimate the effect of landlord size on coarse and left-digit pricing, after including unit fixed effects in our main specification. Thus, all variation in landlord size comes from variation within the exact same unit that has its price set by different landlords before and after a sale.²⁷ These specifications also include neighborhood-by-quarter fixed effects, as well as detailed controls for the time between leases and the seasonal timing of both the current and previous lease. Thus, this comparison holds fixed unobserved, time-invariant, unit-level heterogeneity, as well as local market conditions, seasonality, and the length of the reset gap.

The results are shown in Table 5. After including unit fixed effects, and thus only using variation in landlord size coming from property sales, we find the exact same qualitative pattern: larger landlords are less likely to set rents at multiples of \$50 or \$100, and are more likely to left-digit price. We estimate elasticities that are slightly attenuated relative to our baseline estimates, but overall quite similar. As a result, we find strong evidence that the patterns of coarse and left-digit pricing we document reflect the causal effect of landlord heterogeneity, rather than other types of unobserved differences across the rental market.

A natural interpretation of these results is that large, corporate landlords employ more systematic pricing systems, whereas small individual landlords rely on coarser heuristics. The co-existence of extensive bunching among small landlords and pervasive left-digit pricing among large and corporate landlords therefore points to heterogeneity in pricing sophistication and optimization behavior—not demand-side preferences—as the primary driver of rent-setting patterns. Our results thus suggest that landlord heterogeneity is an important determinant of rent-setting behavior. An implication of this result is that landlords have significant market power: Landlords are able to set systematically different rents and still successfully fill units. Our results thus provide new evidence on the extent of market power in the rental housing market (Watson and Ziv, 2024; Baker, 2025).

While the AHS does not include information on landlords that would allow us to directly

²⁷In this specification, we also use the maximum number of rental units owned at any point by an individual landlord to ensure that this is the only source of variation in landlord size. However, the results are very similar if we use the time-varying measure.

compare our Berkeley estimates to the broader U.S. population of owners, we find similar evidence of coarse pricing in the national AHS sample of renter households. Appendix Figure C.2, Panel (A) reveals substantial bunching in the nominal rent distribution, though somewhat less pronounced than in Berkeley. In the AHS, approximately 23% of rents are set at multiples of \$100 and 18% at multiples of \$50, but not \$100. These shares exceed those reported by Genesove (2003) using the same dataset for an earlier period, suggesting that round-number bunching has become a pervasive feature of modern rental markets, not one confined to Berkeley.

The differences likely reflect that bunching at round multiples becomes more attractive as nominal rent levels rise. Appendix Figure C.2 Panel (B), which is restricted to 1999-2013, and thus more closely aligned with our Berkeley sample, shows higher rates of bunching: 29% of rents at \$100 multiples and 19% at \$50 multiples. Overall, we find strong evidence of round-number pricing comparable to our results for Berkeley alone, with differences plausibly explained by Berkeley's higher rent levels and the later sample period. Our results suggest that coarse pricing has increased substantially over time and that rising nominal rents may further amplify the prevalence of round-number pricing.

In contrast, we observe substantially less left-digit pricing in the broader AHS sample compared to the Berkeley administrative data. The AHS still exhibits a clear spike at prices exactly \$5 below each \$100 threshold, but that spike is only 17% as large as the spike at the round \$100 itself, versus 42% in Berkeley. These differences may stem from the superior precision of our administrative data, particularly if respondents themselves round to multiples when responding to the survey.

This disparity is unlikely to stem from Berkeley-specific features such as its large student population. Appendix Figure A.9 maps left-digit pricing across neighborhoods. Although we detect considerable bunching in student-dominated areas near downtown and U.C. Berkeley's campus, we also find substantial bunching in neighborhoods with minimal student presence, including southern areas bordering Oakland, West Berkeley, and the affluent Berkeley hills. By design, our dataset excludes much of the housing for undergraduate students: campus dormitories, large shared houses, student co-ops, and Greek residences. Notably, all but one Berkeley census tract contains more bunching than is observed in the AHS. Our analysis underscores the value of contract-level administrative datasets and illuminates the need for additional research utilizing similar rent registries to assess left-digit bunching patterns in the broader rental market.

If coarse pricing patterns reflect less sophisticated optimization, landlords may rely more heavily on these strategies when the pricing environment becomes more complicated or less routine. As a result, we next examine whether round-number bunching and coarse pricing

covary with rental price growth. Once again, we include unit fixed effects and detailed controls for the time between leases to compare pricing behavior for the same unit in more versus less typical rent-growth periods, holding fixed the reset gaps. Figure 6, Panel (A) plots the propensity to set rents at multiples of \$100 as a function of the change in the rental price index. We find evidence of a strong, non-monotonic relationship between the prevalence of heuristic pricing and changes in the rent price index. The propensity to use coarse pricing practices falls when rent growth is near its median value (approximately 4% over the sample period, denoted by the dashed vertical line), and is significantly higher when rents are either falling or rising sharply. In contrast, Panel (B) shows the opposite pattern for left-digit pricing: it is most prevalent during periods of moderate rent growth and declines during unusually large increases or decreases in rents.

This non-monotonic pattern indicates that the precision of rent setting varies with the complexity of the pricing environment. When rent growth is moderate, landlords appear to optimize more precisely by avoiding coarse pricing and strategically employing left-digit pricing. In contrast, when rents are falling or rising rapidly, the deviation from typical nominal adjustment may reduce landlords' capacity or willingness to fine-tune prices, leading to greater reliance on round-number rules. More broadly, our findings point to a potential cost of inflation volatility that is typically absent from standard New Keynesian models: periods of unusual price growth may reduce pricing precision in housing markets. The extent to which this is a feature of other markets is an interesting direction for future research.

Table 6 provides a formal test of this hypothesis. We estimate how pricing precision covaries with rental price growth, allowing for different slopes above and below median price growth during our sample period.²⁸ Panel (A) shows the overall results for all landlords. Across all three measures of pricing imprecision, we find that periods of unusual price changes lead to more bunching at round numbers and less left-digit pricing, suggesting that inflation volatility reduces the precision of rental pricing. We find that the propensity to bunch at multiples of \$50 or \$100 increases sharply as rental prices increase above its median level. We also estimate marginally significant slopes that are about half as large in magnitude as rental price growth falls below its median. In contrast, for left-digit pricing we find especially large reductions in pricing precision as rent growth falls below its usual levels. Overall, we find strong evidence that pricing precision falls as rent growth deviates from its usual values, suggesting that the complexity of the pricing environment reduces the precision with which landlords set prices.

We further validate our interpretation that unusual pricing environments reduce pricing precision using an alternative source of variation in the magnitude of optimal rent adjustment—

²⁸We also include unit fixed effects, thus estimating whether pricing precision falls within the exact same unit when it is transacted in more vs. less normal times for rent price growth.

the time elapsed between leases. Appendix Figure A.10 shows that larger price changes associated with longer gaps between contracts are linked to coarser pricing patterns. Panel (A) shows that when the interval between leases is longer, landlords are substantially more likely to set rents at round numbers. Panel (B) shows the mirror pattern for left-digit pricing. These results provide direct evidence for our proposed mechanism: larger nominal price changes reduce the precision of rent setting, as reflected in the shift from fine-tuned to coarser pricing practices.

Consistent with our results on heterogeneous landlord sophistication, we find that pricing sophistication deteriorates most for smaller landlords as the pricing environment gets more complex. Figure 6, Panels (C) and (D) show how coarse pricing patterns respond to business cycle variation separately for large and small landlords. Overall, the patterns are strongest for smaller landlords. The coefficients for small landlords are similar to the coefficients for the full sample. However, we also find similar qualitative patterns for larger landlords, especially as prices fall below their median, where the magnitudes are larger than for small landlords. Table 6, Panel (B) shows the effects after residualizing on unit fixed effects and the time between leases, separately for small and large landlords. Overall, we find that the results for small landlords are similar to the results for the overall sample. In contrast, for large landlords, we only find evidence that pricing precision deteriorates as prices fall below their usual levels. This suggests that while cognitive constraints triggered by volatile inflation are larger for small, less sophisticated firms, they also affect larger firms, especially when inflation is negative.

We find similar results for large versus small landlords when using lease contract gaps as a source of variation in the magnitude of ideal price adjustments. Appendix Figure A.10, Panels (C) and (D) show how pricing precision varies with the time between leases. We find that as time between leases grows, and thus the size of the rent adjustment increases, the propensity to set rents at multiples of \$100 increases for both landlord types. However, the semi-elasticity is more than 3 times larger for small landlords than for large landlords, suggesting that pricing precision deteriorates more for small landlords than for large landlords as the difficulty of the pricing problem increases. Similarly, we find that the propensity to left-digit price falls with the time between leases, but only for small landlords. As a result, we find evidence that pricing precision deteriorates especially strongly for small landlords as the pricing environment gets more complex, consistent with our evidence resulting from variation over the business cycle.

Overall, our heterogeneity results are consistent with our interpretation that cognitive costs drive the response of pricing precision to volatile inflation and large nominal adjustments. We find the results are primarily driven by smaller, less sophisticated landlords, though we do still find significant effects for larger landlords, especially when inflation is negative.

4.2 Fact 5: Heterogeneous Adjustment Over the Business Cycle

Fact 4 showed that landlord heterogeneity is a powerful predictor of rent setting behavior. We next ask whether or not landlords also differ in how they adjust rents over the business cycle. Our final fact is that larger landlords adjust rents more aggressively in response to changing market conditions. Table 7, Panel (A) tests how landlord size affects how rents change at tenant turnover. Columns (1) through (3) indicate that, on average, large landlords are less likely to leave rents unchanged, but are more likely to reduce rents when a new tenant moves in. Column (4) shows that large landlords are also more likely to implement large rent increases of more than 20 log points at turnover. As a result, the average rent change at turnover is larger for large landlords (Column (5)). These regressions include detailed controls for the time between leases and neighborhood-by-quarter fixed effects, ensuring that the results are not driven by market-level differences or variation in the optimal reset price. Because larger landlords are more likely to decrease rents, and less likely to leave rents unchanged, our results suggest that large landlords are less bound by the nominal rigidities documented in Section 3.1. However, because rents set by larger landlords increase more on average, our results suggest that increased ownership of rental housing by large landlords could lead to non-trivial price increases.

These findings provide further evidence that cross-landlord heterogeneity reflects differences in sophistication and optimization behavior rather than tenant demand. While tenants may prefer salient or round-number rents, it is implausible that they hold systematic preferences over the change in rent between leases, especially since a new tenant occupies the unit after turnover. Because different landlords have differing responses to identical opportunities to reset rents suggests that the observed heterogeneity in rent-setting behavior primarily reflects variation in pricing sophistication and optimization, rather than differences in tenant composition or demand.

Table 7, Panel (B) shows that heterogeneity in rent changes at tenant turnover is largely driven by differential cyclical responsiveness. The table reports regressions of rent-change indicators on landlord size interacted with terciles of average rent growth in Berkeley. These regressions include neighborhood-by-quarter fixed effects, current-by-previous-lease-start-month fixed effects, and detailed controls for the time between leases. Thus, we compare landlords setting rents in the same local market and with the same seasonal lease-timing pattern, while holding fixed differences in the length of the reset gap.

Column (1) shows that while, on average, large landlords increase rents at similar rates as small landlords, there is significant heterogeneity over the business cycle. The negative coefficient in the first row indicates that during the weakest phases of the rental cycle, large landlords are significantly less likely to increase rents. In contrast, during high-growth periods,

large landlords are more likely than small landlords to increase rents.

Column (2) shows an analogous pattern for rent decreases: large landlords are much more likely to cut rents during downturns, and are slightly less likely to lower rents during booms. Finally, Column (3) indicates that larger landlords are less likely to leave rents unchanged during downturns, suggesting that they adjust prices more actively when market conditions deteriorate. These results also suggest that the absence of binding downward nominal rent rigidity in downturns is largely driven by the behavior of large landlords, who are more willing to cut rents and less likely to leave rents unchanged. Overall, our results indicate that larger landlords adjust more strongly on the extensive margin in response to the business cycle, whereas the extensive-margin rent-setting decisions of small landlords are less responsive to market conditions.

Column (4) shows that this same pattern exists for the size of rent increases. Column (4) shows that large landlords are no more likely than small landlords to make large rent increases (exceeding 20 log points) during busts, but are much more likely to do so during periods of especially strong rent growth. Column (5) summarizes the aggregate effect: in downturns, rents set by large landlords change by about the same amount as those set by small landlords, increase by slightly more in normal times, and increase substantially more during booms. Taken together, these findings indicate that large landlords are significantly more responsive to the business cycle than smaller landlords, both in the likelihood of adjusting rents and in the magnitude of their adjustments.

To provide a further test of whether these relationships reflect causal effects of landlord heterogeneity, we again turn to our sample of sold units. We estimate the exact same regression as our baseline, but include unit fixed effects.²⁹ Thus, this regression tests whether landlords of different sizes respond differently to the business cycle when setting rents *for the exact same unit*.

Table 8 shows the results in this ever sold sample. As with Fact 4, the effects in the regression with unit fixed effects are very similar to the baseline. Though somewhat less precisely estimated, we find strong evidence that large landlords are more likely to raise rents and less likely to reduce rents in boom markets, and more likely to cut rents in the lowest growth markets. Similarly, large landlords are also more likely to give large rent increases of more than 20 log points in the highest growth markets, and on average raise rents by more in boom times. Overall, we find very similar qualitative and quantitative results in both our overall sample, and when using quasi-experimental variation from property sales. Taken together, our results provide strong evidence for a causal effect of landlord heterogeneity on the cyclicity of rent

²⁹As with the previous regression, we only include variation in landlord size from sales by using the maximum number of units ever owned by a landlord as the independent variable.

setting over the business cycle.

These cyclical patterns bolster our interpretation that cross-landlord differences in pricing reflect variation in sophistication and optimization behavior and highlight the broader role of firm heterogeneity in determining inflation dynamics. Larger landlords appear more attentive to the business cycle: in downturns, they reoptimize by cutting or adjusting rents rather than leaving prices unchanged, while in expansions they implement larger rent increases. These results also imply that a shift in market composition toward larger, more professional landlords would increase the volatility of rental prices over the cycle. By amplifying the sensitivity of rents to aggregate conditions, such compositional shifts may in turn influence the persistence and volatility of shelter inflation.

4.3 Discussion

Our final two facts highlight the role of landlord heterogeneity in price setting, both on average and in response to shocks. The facts presented in Section 3 suggested that housing markets are best modeled by a Calvo, time-dependent pricing friction combined with a fixed cost that landlords had to pay to actively reoptimize. The facts presented in this section suggest that this second cost comes from managerial or cognitive costs and varies across different landlords.

In particular, our results suggest that large landlords respond more strongly to market conditions. We show that larger landlords are less bound by nominal rent rigidities and are more responsive to changing business cycle conditions. Larger landlords also engage in more sophisticated pricing strategies, such as setting rents just below multiples of \$100. In contrast, small landlords are less responsive to the business cycle and often use coarse pricing. We also find that as the pricing environment gets more complicated, as a result of unusual changes in market conditions or larger price gaps driven by longer time between leases, the use of sophisticated pricing strategies (left-digit pricing) deteriorates, and the use of pricing heuristics (multiples of \$50 or \$100) increases. This reduction in precision is especially large for smaller landlords, though large landlords also show similar patterns, especially when rent growth is below its usual levels. Thus, we document a novel cost of inflation, where volatility in inflation causes a reduction in pricing precision.

These results imply that larger landlords amplify the response of shelter prices to shocks, and that increasing the share of large landlords would increase the volatility of shelter prices over the business cycle. Our results thus complement recent work by Calder-Wang and Kim (2024) who find algorithmic pricing makes rents more responsive to economic shocks.

5 How Do Large Landlords Affect Shelter Inflation?

Taken together, our empirical results point to a rent-setting process in which tenant turnover creates opportunities to reset rents, but landlords do not always fully reoptimize, and the frictions governing adjustment differ across landlords. We next formalize these features in a model of rental price setting and use the model to quantify how landlord heterogeneity affects rents for new tenants. In this section, we introduce and estimate our model using moments from our pricing data, then employ the model to quantify how the frictions documented in our empirical section shape shelter inflation.

5.1 Model Summary

We use the new empirical facts to discipline a model of rental price setting. Our five facts point to a hybrid price-setting environment that combines features of both time- and state-dependent models. Acyclical tenant turnover provides landlords with the opportunity to reset rents, as in a Calvo (1983) model, but landlords who receive this opportunity do not always reoptimize. Instead, active reoptimization requires attention, information processing, and managerial effort, which we model as a fixed cost. Posting a nominal rent cut is subject to an additional fixed cost. We allow for heterogeneity in costs between large and small landlords.

When a tenant chooses to leave a unit, the landlord can either keep the original rent or actively choose a new rent. Over time, prices evolve as a result of both changes in market conditions and idiosyncratic changes in demand for individual units. Let the full-information change in the frictionless log rent for unit i between time t and the start of the previous lease be given by:

$$\Delta m_{i,t} = M_{i,t} + U_{i,t} \quad (1)$$

where $M_{i,t}$ is the change in desired rents due to changes in market conditions and $U_{i,t}$ is the idiosyncratic, unit-level change. Landlords have imperfect information about how demand for their unit has evolved over time, and instead observe noisy signals of both the change in both market conditions and the due to unit-specific factors. We allow the precision of these signals to vary for small and large landlords, indexed by $g \in \{S, L\}$. Thus, the signals received by the landlords are given by:

$$S_{i,t,g}^M = M_{i,t} + v_{i,t,g}^M, \quad S_{i,t,g}^U = U_{i,t} + v_{i,t,g}^U.$$

Here, $v_{i,t,g}^M$ and $v_{i,t,g}^U$ represent the noisiness of the signals received by landlords. Then, landlords form posteriors about the frictionless market price for their unit:

$$\widetilde{\Delta m}_{i,t,g} = E[\Delta m_{i,t} | S_{i,t,g}^M, S_{i,t,g}^U].$$

Because landlords of different sizes may receive higher or lower quality information about market- or unit-level conditions, they will respond differently to information about both the market and idiosyncratic demand. In estimation, we use our repeat-rent index as a proxy for the market innovation. As a result, the desired new-tenant rent change for a landlord of type g at tenant turnover will be given by:

$$x_{i,t,g} = b_{i,t-1,g} + \theta_g A_{i,t} + \sigma_{z,g} \sqrt{L_{i,t}} \epsilon_{i,t}.$$

In this formulation, $b_{i,t-1,g}$ represents any gap between rents set by landlords and the $x_{i,t,g}$ that is carried over from a previous lease,³⁰ $A_{i,t}$ represents the cumulative change in the repeat-rent index between the current lease and previous lease, and $\sqrt{L_{i,t}} \epsilon_{i,t}$, with $\epsilon \sim N(0, 1)$ represents the cumulated, standardized, unit-specific innovation since the previous lease, which grows with the time between leases. The parameters to be estimated are θ_g , which measures the responsiveness of different landlords to market-specific information, and $\sigma_{z,g}$, which measures the responsiveness of different landlords to unit-specific information.

Once the landlord has a target rent, she balances potential losses from mispricing her unit with the costs associated with actively resetting her rent. At turnover, the landlord draws a reoptimization cost, which takes the following form (Blanco et al., 2025):

$$K_{i,t,g}^r = \begin{cases} 0, & \text{with probability } 1 - \lambda_g^r, \\ \kappa_g^r U_{i,t}^r, & \text{with probability } \lambda_g^r, \end{cases} \quad U_{i,t}^r \sim U[0, 1] \quad (2)$$

In addition, landlords draw an additional cost that they pay when reducing rents below their previous value. This rent-cut cost takes the following form:

$$K_{i,t,g}^c = \kappa_g^c U_{i,t}^c, \quad U_{i,t}^c \sim U[0, 1] \quad (3)$$

Thus, the landlord pays the fixed costs and updates the rent when $\frac{1}{2}x_{i,t,g}^2 \geq K_{i,t,g}^r + \mathbf{1}\{x_{i,t,g} < 0\}K_{i,t,g}^c$.³¹ When this condition holds, the landlord fully closes the gap between the incumbent rent and the posterior for the frictionless rent. When the fixed costs are greater than the cost of mispricing, the landlord keeps the rent at its previous value.³²

³⁰In estimation, we initialize this rent gap to zero for the first lease.

³¹This expression results from the fact that the profit loss from mispricing are approximately quadratic in the deviation of prices from the optimum.

³²We assume landlords are myopic and solve a static problem at each opportunity created by tenant turnover. As a result, landlords do not take into account that not updating their rents today will lead to larger price gaps

We estimate our model on the Berkeley rental data using simulated method of moments (SMM). The set of parameters to be estimated are given by: $\{\theta_g, \sigma_{z,g}, \kappa_g^c, \kappa_g^r, \lambda_g^r\}$. We take our rental data panel as given, including information on landlord size.³³ For each candidate parameter vector, we create an expected rent (and thus rent changes) for each observation. We estimate the model using the TikTak global solution algorithm described in Arnoud, Guvenen and Kleineberg (2022). We target 16 moments of the data: 6 cross-sectional moments of the rent-change distribution, 4 moments that describe how the probability of actively changing rents varies with time between leases, and 6 moments that parameterize differences between large and small landlords.

The model matches a variety of targeted and untargeted moments. Table 9 shows the estimated parameters, as well as our targeted moments. Consistent with our empirical estimates, we find that larger landlords respond more strongly to information about market conditions than small landlords, while small landlords respond more strongly to idiosyncratic, unit-level information. This is consistent with our finding that large landlords are more responsive to the business cycle. In addition, we also find that small landlords pay large costs of cutting rents, while the estimated parameter for large landlords is close to zero. This, again, is consistent with our empirical finding that downward nominal rent rigidity is mostly driven by small landlords. In contrast, both parameters governing the distribution of the fixed cost to actively reoptimize are larger for large landlords than for small landlords. As a result, we find that small landlords are more likely to actively reoptimize when given an opportunity as a result of tenant turnover. Note that while this seems to contradict our results that large landlords are less likely to leave rents unchanged between leases, in the model rents staying fixed can result from either not actively reoptimizing or from a hesitancy to cut rents, which we estimate is much larger for small landlords. In addition, the model is able to hit the overall difference between large and small landlords in the propensity to leave rents unchanged, as shown in Table 9, Panel (B).

Table 9, Panel (B) shows that our model is able to closely hit most of our targeted moments in estimation. Figure 7 replicates Figure 2 in both the actual data and in data generated by our estimated model. We nearly exactly hit the number of rents that do not change in the cross-section, the number of rent cuts, and the observed differences across landlords. Relative to the data, our model slightly underpredicts overall dispersion in the log rent change distribution as well as the number of small rent increases. Thus, our model inherits the well-known pathology of many menu-cost models that Ss-style decision rules are not able to match the number of

when future tenants leave.

³³The panel used in estimation is slightly smaller than the data we use in our empirical analysis because we require all observations to have data on the previous tenant's initial rent. However, as observed in Section 5, the coefficients and moments are nearly identical to those in the full dataset.

small price increases that we see in the data. Additionally, the semi-elasticity of rent rigidity with respect to time between leases is too flat in the model relative to the data. However, on average our model is able to capture salient features of the rent-change distribution.

In addition, despite not targeting these moments in estimation, our model also closely captures the dynamics of rental prices over the business cycle. Appendix Table B.3 replicates Table 3 in the model-generated data. We nearly exactly capture the dynamics of how rents adjust over the business cycle on both the extensive and intensive margins. Similarly, Table 10 shows that we closely capture how landlord heterogeneity affects business cycle dynamics. In our model, just as in the data, large landlords are more likely to cut rents in downturns and more likely to raise rents in booms. In addition, in the model, large landlords are more likely to raise rents by large amounts in very good market conditions, and thus overall rent changes are larger for large landlords in expanding markets. Our model-generated results match the data both qualitatively and quantitatively, despite the fact that these moments were not explicitly targeted in estimation. These results provide additional confidence that despite the relative simplicity of our model, it is able to closely capture how different landlords change rents both in the cross section, and dynamically over the business cycle.

5.2 Estimated Impact of Large Landlords on Shelter Inflation

We use our estimated model to quantify the effect of large landlords on rental price inflation. A large academic and public debate has centered on the role institutional landlords play in the rental market. For example, recent Congressional and Executive actions have sought to limit the ability of large institutional investors to purchase single-family homes.

We use our quantitative model to study how the entry of large landlords would affect rents for new tenants.³⁴ We consider two counterfactuals: one in which we double the share of large landlords and one in which we replace all small landlords with large landlords.³⁵ In our estimated model, landlords differ in the frictions and information they have when setting prices. Thus, our counterfactual is equivalent to giving small landlords access to the same cost distribution and information set as large landlords. We hold all other aspects of the environment fixed.

Figure 8 shows the effect of changing the number of large landlords on rental prices for new tenants. Unsurprisingly given our empirical results, we find that increasing the share of

³⁴Our focus is slightly different from the policy debate in that our definition of large landlord is based on whether or not the owner rents 75 or more units. Our evidence thus speaks to a broader segment of the rental market than currently targeted by proposed policy actions, although given that many of our results seem to be linear (in logs) in the number of units a landlord owns, our results would be even larger if we focused on the largest, institutional landlords

³⁵In our data, about 33% of leases are in units owned by large landlords.

large landlords significantly amplifies the volatility of rental prices. In the largest boom years of the rental market, associated with the tech booms in the early 2000s and early 2010s, we find that replacing small landlords with large landlords would have increased rents for new tenants by more than 1 percentage point. In addition, during the period of rent growth before the Great Recession, new tenant rent growth would be almost 0.75 percentage points higher.

In contrast, we find that in economic downturns, such as the 2000s dot-com bust and the Great Recession, rents would have fallen by slightly more, though the effects are modest. Over the entire sample, the average effect is positive, suggesting that increasing the share of large landlords would significantly increase rental price growth. Appendix Table B.4 shows that, on average, we estimate that replacing all small landlords with large landlords would increase the growth of new tenant rents by 0.40 percentage points per year. It also increases the time-series volatility of annual new-tenant rent inflation. In high-growth periods new tenant rents grow by about 1 p.p. more per year than in the current status quo, while in low-growth periods they are lower by about 0.06 p.p. Overall, our results suggest that replacing small landlords with large landlords would increase shelter price volatility and increase the average level of shelter inflation. However, we stress that these counterfactuals are inherently partial-equilibrium in nature, in that we do not model the endogenous response of households' search behavior or potential changes in rental housing supply that might be caused by larger average price growth, or greater variance in price changes. We think incorporating these channels is an important avenue for future work.

6 Conclusion

In this paper, we use detailed administrative data from a single city and nationally representative survey data to document new facts about rental price setting in the United States. Our results provide new information on how shelter prices adjust over time, which is crucial for understanding both aggregate inflation and the cost of living. Our work also provides new empirical moments that can be used more broadly for calibrating models with time-dependent pricing frictions.

We highlight two novel implications of our results. First, we provide new evidence that firm heterogeneity is an important determinant of pricing decisions and inflation dynamics. We find evidence that more sophisticated landlords are more responsive to the business cycle and less bound by nominal rent rigidities. This suggests that firm heterogeneity and structural transformation, including the rise of large, superstar firms, are important determinants of inflation dynamics. Second, we document that unusually volatile inflation and large nominal price adjustments leads firms to use coarser pricing strategies. This suggests a novel cost of inflation,

whereby unusually volatile inflation reduces pricing precision. Future work could study the extent to which these trends are common more broadly outside of the housing sector.

Finally, our work provides guidance for modeling price-setting in macroeconomic models with housing. We find that rental pricing exhibits distinct dynamics from other sectors of the economy. Our results suggest that pricing frictions in the housing market are most consistent with time-dependent Calvo models. Interestingly, this finding holds even in the broader U.S. market, which suggests that a combination of long-term rental contracts and rent rigidity among lease renewals makes new tenant pricing more akin to Calvo than menu-cost models (Adams et al., 2024; Ball and Koh, 2025). However, we also find that Calvo frictions alone cannot explain pricing patterns in the shelter sector. Instead, a combination of time-dependent frictions and a cost of reoptimizing is necessary to fit the empirical patterns in the data, including the large degree of rigidity in the rental housing market, even during periods when new tenants enter a unit. Our findings suggest that these reoptimization costs stem from cognitive or managerial frictions in price adjustment, and that these costs vary significantly across different firms.

Our results have important implications for public policy. We contribute to a large debate about the welfare implications of large landlords. We find that allowing large landlords to take over the housing stock currently owned by smaller landlords would increase average rental inflation for new tenants by about 0.40 percentage points per year, with especially large effects in periods of rapid price growth, when prices would increase by about 1 p.p. more than in the current baseline. However, the ultimate welfare effects of these changes, after incorporating endogenous, general equilibrium responses are uncertain. More responsive price setting might allow for a more efficient allocation of housing and, if it increases profits, might increase rental housing supply. However, differences in pricing strategies across landlords might also cause a reallocation between consumer and producer surplus, especially as our results may suggest that landlords have substantial pricing power in the rental housing market. Future work could disentangle the overall welfare impacts of our findings for consumers and producers.

References

- Adams, Brian, Lara Loewenstein, Hugh Montag, and Randal Verbrugge.** 2024. “Disentangling Rent Index Differences: Data, Methods, and Scope.” *American Economic Review: Insights*, 6(2): 230–245.
- Alvarez, Fernando, Francesco Lippi, and Aleksei Oskolkov.** 2021. “The Macroeconomics of Sticky Prices with Generalized Hazard Functions.” *The Quarterly Journal of Economics*, 137(2): 989–1038.
- Amiti, Mary, Oleg Itskhoki, and Jozef Konings.** 2019. “International Shocks, Variable Markups, and Domestic Prices.” *The Review of Economic Studies*, 86(6): 2356–2402.
- Anderson, Eric T., and Duncan I. Simester.** 2003. “Effects of \$9 Price Endings on Retail Sales: Evidence from Field Experiments.” *Quantitative Marketing and Economics*, 1(1): 93–110.
- Arnoud, Antoine, Fatih Guvenen, and Tatjana Kleineberg.** 2022. “Benchmarking Global Optimizers.” *Working paper*, November 23, 2022.
- Aysoy, Cem, Cevriye Aysoy, and Semih Tumen.** 2014. “Quantifying and Explaining Stickiness in Housing Rents: A Turkish Case Study with Micro-Level Data.” *Journal of Housing Economics*, 25: 62–74.
- Baker, Sarah.** 2025. “Property Tax Pass-Through to Renters: A Quasi-experimental Approach.”
- Ball, Laurence M., and Kyung Woong Koh.** 2025. “Market Rents and CPI Shelter Inflation.” National Bureau of Economic Research Working Paper 34113.
- Barbieri, Felipe, and Gregory Dobbels.** 2025. “Market Power and the Welfare Effects of Institutional Landlords.” Working Paper.
- Barsky, Robert B., and Jeffrey A. Miron.** 1989. “The Seasonal Cycle and the Business Cycle.” *Journal of Political Economy*, 97(3): 503–534.
- Bhattarai, Saroj, and Raphael Schoenle.** 2014. “Multiproduct Firms and Price-Setting: Theory and Evidence from U.S. Producer Prices.” *Journal of Monetary Economics*, 66: 178–192.
- Bianchi, Javier, Alisdair McKay, and Neil Mehrotra.** 2024. “How Should Monetary Policy Respond to Housing Inflation?” Federal Reserve Bank of Minneapolis.
- Bils, Mark, and Peter J. Klenow.** 2004. “Some Evidence on the Importance of Sticky Prices.” *Journal of Political Economy*, 112(5): 947–985.

- Blanco, Andres, Corina Boar, Callum Jones, and Virgiliu Midrigan.** 2025. “Nonlinear Inflation Dynamics in Menu Cost Economies.” Working Paper.
- Calder-Wang, Sophie, and Gi Heung Kim.** 2024. “Algorithmic Pricing in Multifamily Rentals: Efficiency Gains or Price Coordination?” Working Paper.
- Calvo, Guillermo A.** 1983. “Staggered Prices in a Utility-Maximizing Framework.” *Journal of Monetary Economics*, 12(3): 383–398.
- Chang, Konhee.** 2025. “Diversifying the Suburbs: Rental Supply and Spatial Inequality.” Working Paper.
- Coven, Joshua.** 2025. “The Impact of Institutional Investors on Homeownership and Neighborhood Access.” Working Paper.
- Davis, Steven J., and Pawel M. Krolkowski.** 2025. “Sticky Wages on the Layoff Margin.” *American Economic Review*, 115(2): 491–524.
- DellaVigna, Stefano, and Matthew Gentzkow.** 2019. “Uniform Pricing in U.S. Retail Chains.” *The Quarterly Journal of Economics*, 134(4): 2011–2084.
- Dube, Arindrajit, Alan Manning, and Suresh Naidu.** 2025. “Monopsony and Employer Misoptimization Explain Why Wages Bunch at Round Numbers.” *American Economic Review*, 115(8): 2689–2721.
- Gagliardone, Luca, Mark Gertler, Simone Lenzu, and Joris Tielens.** 2025. “Micro and Macro Cost-Price Dynamics in Normal Times and During Inflation Surges.” Working paper.
- Gallin, Josh, Lara Loewenstein, Hugh Montag, and Randal Verbrugge.** 2025. “Rent Setting for New and Continuing Tenants.”
- Gallin, Joshua, and Randal J Verbrugge.** 2019. “A Theory of Sticky Rents: Search and Bargaining with Incomplete Information.” *Journal of Economic Theory*, 183: 478–519.
- Genesove, David.** 2003. “The Nominal Rigidity of Apartment Rents.” *Review of Economics and Statistics*, 85(4): 844–853.
- Genesove, David, and Christopher Mayer.** 2001. “Loss Aversion and Seller Behavior: Evidence from the Housing Market.” *The Quarterly Journal of Economics*, 116(4): 1233–1260.
- Golosov, Mikhail, and Robert E. Lucas Jr.** 2007. “Menu Costs and Phillips Curves.” *Journal of Political Economy*, 115(2): 171–199.

- Gorback, Caitlin, Franklin Qian, and Zipei Zhu.** 2025. “The Impact of Institutional Owners on Housing Markets.” Working Paper.
- Gurun, Umit G, Jiabin Wu, Steven Chong Xiao, and Serena Wenjing Xiao.** 2022. “Do Wall Street Landlords Undermine Renters’ Welfare?” *The Review of Financial Studies*, 36(1): 70–121.
- Hanson, Sebastian.** 2024. “Institutional Investors in the Market for Single-Family Housing: Where Did They Come From, Where Did They Go?” Working Paper.
- Harwood, Katharine, Ingrid Gould Ellen, and Katherine O’Regan.** 2025. “The Rise of Corporate Landlords: An Examination of Behavioral Differences in the Multi-Family Market.”
- Hazell, Joe, and Bledi Taska.** 2025. “Downward Rigidity in the Wage for New Hires.”
- Hoffmann, Johannes, and Jeong-Ryeol Kurz-Kim.** 2006. “Consumer Price Adjustment Under the Microscope: Germany in a Period of Low Inflation.”
- Hosken, Daniel, and David Reiffen.** 2004. “Patterns of Retail Price Variation.” *The RAND Journal of Economics*, 35(1): 128–146.
- Klenow, Peter J., and Benjamin A. Malin.** 2010. “Microeconomic Evidence on Price-Setting.” In . Vol. 3 of *Handbook of Monetary Economics*, , ed. Benjamin M. Friedman and Michael Woodford, 231–284. Elsevier.
- Klenow, Peter J., and Oleksiy Kryvtsov.** 2008. “State-Dependent or Time-Dependent Pricing: Does it Matter for Recent U.S. Inflation?” *The Quarterly Journal of Economics*, 123(3): 863–904.
- Lacetera, Nicola, Devin G. Pope, and Justin R. Sydnor.** 2012. “Heuristic Thinking and Limited Attention in the Car Market.” *American Economic Review*, 102(5): 2206–36.
- Lambie-Hanson, Lauren, Wenli Li, and Michael Slonkosky.** 2022. “Real Estate Investors and the U.S. Housing Recovery.” *Real Estate Economics*, 50(6): 1425–1461.
- Matějka, Filip.** 2016. “Rationally Inattentive Seller: Sales and Discrete Pricing.” *The Review of Economic Studies*, 83(3): 1125–1155.
- Molloy, Raven.** 2024. “Discussion of: *America’s Housing Affordability Crisis and the Decline of Housing Supply* by Edward Glaeser and Joseph Gyourko.”

- Mondragon, John, Addie New-Schmidt, Jordan Rosenthal-Kay, Johannes Wieland, and Caleb Wroblewski.** 2026. “Does a Rising Tide Lift all Boats? Evidence from the Bay Area Tech Boom.” Working Paper.
- Montag, Hugh, and Daniel Villar.** 2023. “Price-Setting During the Covid Era.” FEDS Notes, Board of Governors of the Federal Reserve System.
- Nakamura, Emi, and Jón Steinsson.** 2008. “Five Facts about Prices: A Reevaluation of Menu Cost Models.” *The Quarterly Journal of Economics*, 123(4): 1415–1464.
- Nakamura, Emi, and Jón Steinsson.** 2010. “Monetary Non-Neutrality in a Multisector Menu Cost Model.” *The Quarterly Journal of Economics*, 125(3): 961–1013.
- Nakamura, Emi, Jón Steinsson, Patrick Sun, and Daniel Villar.** 2018. “The Elusive Costs of Inflation: Price Dispersion during the U.S. Great Inflation.” *The Quarterly Journal of Economics*, 133(4): 1933–1980.
- Neroli, Austin.** 2022. “Keeping Up with the Blackstones: Institutional Investors and Gentrification.” Working Paper.
- Ngai, L. Rachel, and Silvana Tenreyro.** 2014. “Hot and Cold Seasons in the Housing Market.” *American Economic Review*, 104(12): 3991–4026.
- Orchard, Jacob.** 2025. “Non-homothetic Demand Shifts and Inflation Inequality.” Federal Reserve Board Working paper. This version: August 2025. First version: February 2021.
- Park, Seongjin.** 2024. “Adaptive Rent Pricing and Its Consequences.” Working Paper.
- Raymond, Lindsey.** 2025. “The Market Effects of Algorithms.” Working Paper.
- Repetto, Luca, and Alex Solís.** 2019. “The Price of Inattention: Evidence from the Swedish Housing Market.” *Journal of the European Economic Association*, 18(6): 3261–3304.
- Shimizu, Chihiro, Kiyohiko G Nishimura, and Tsutomu Watanabe.** 2010. “Residential Rents and Price Rigidity: Micro Structure and Macro Consequences.” *Journal of the Japanese and International Economies*, 24(2): 282–299.
- Stevens, Luminita.** 2020. “Coarse Pricing Policies.” *The Review of Economic Studies*, 87(1): 420–453.
- Strulov-Shlain, Avner.** 2022. “More Than a Penny’s Worth: Left-Digit Bias and Firm Pricing.” *The Review of Economic Studies*, 90(5): 2612–2645.

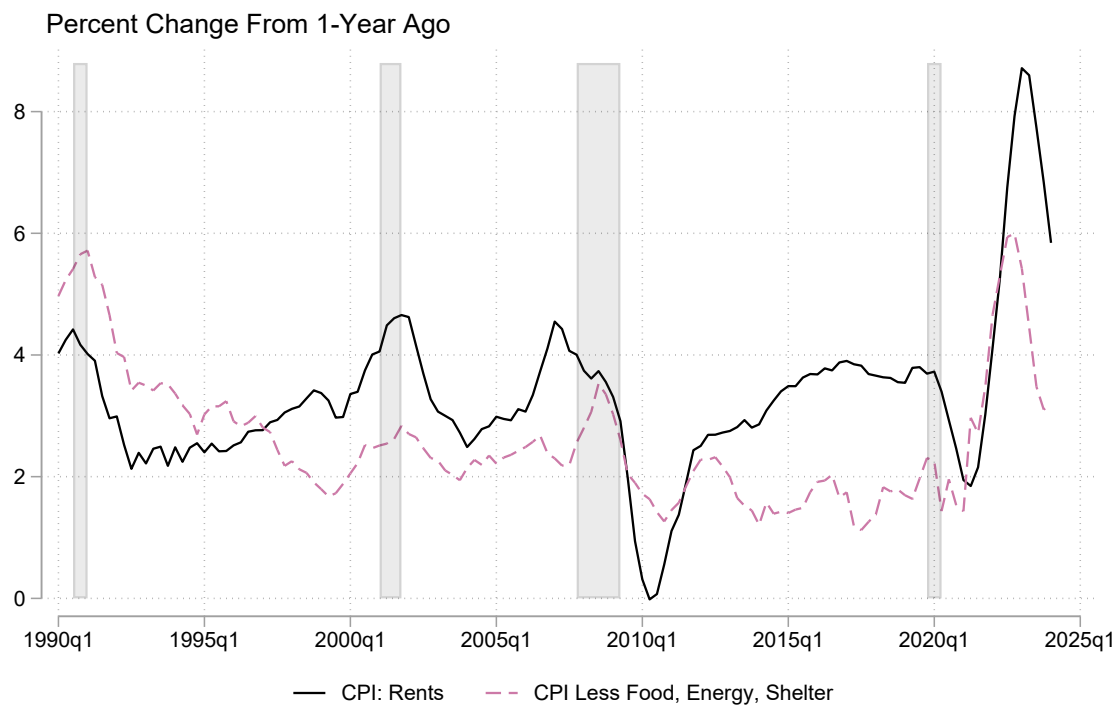
Strulov-Shlain, Avner. 2024. “Firms Have Partial Knowledge and they Partially Optimize: Evidence From a Reform.”

Watson, C Luke, and Oren Ziv. 2024. “A Test for Pricing Power in Urban Housing Markets.”

Yang, Choongryul. 2022. “Rational Inattention, Menu Costs, and Multi-Product Firms: Micro Evidence and Aggregate Implications.” *Journal of Monetary Economics*, 128: 105–123.

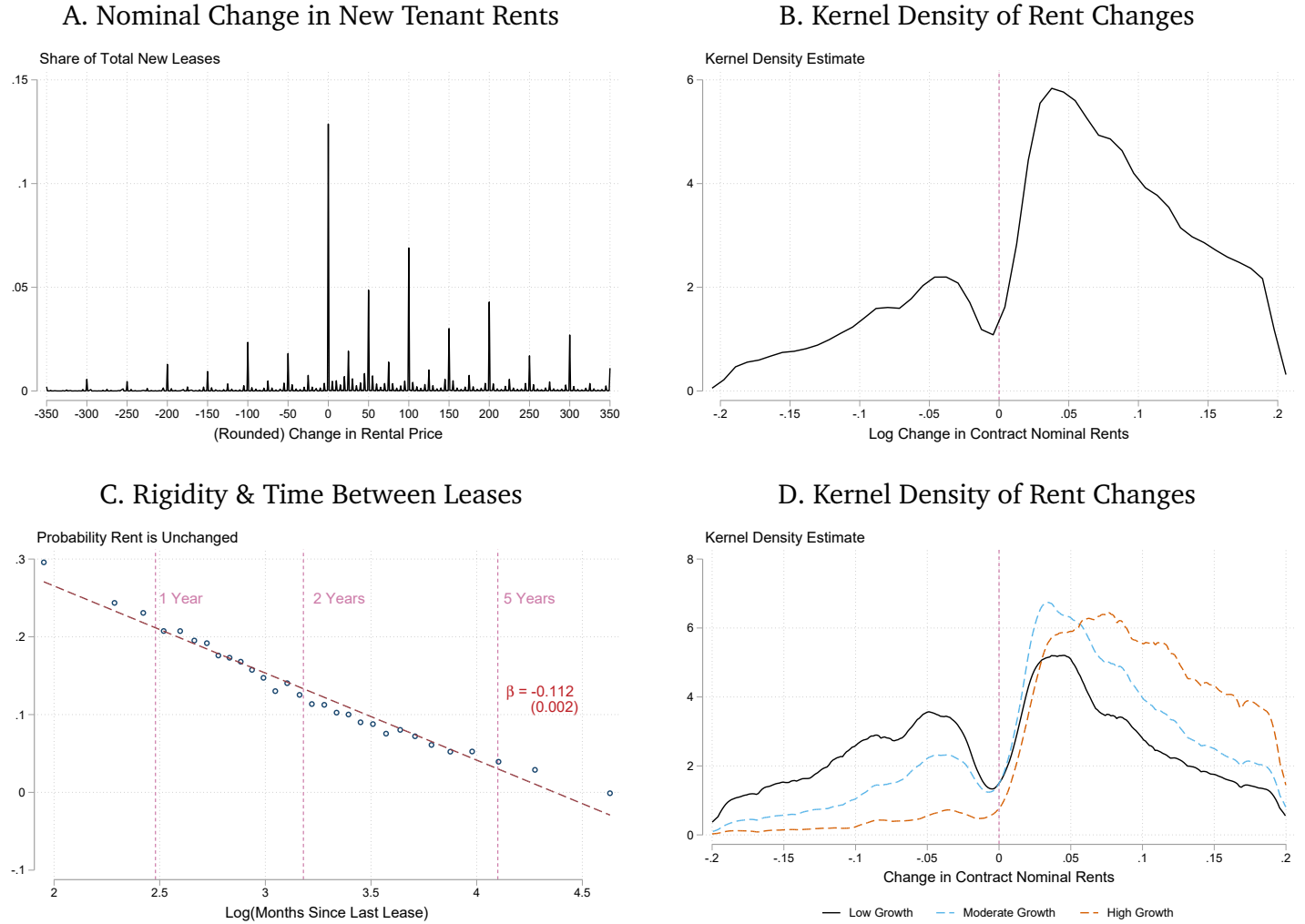
7 Figures

Figure 1: Time Series of Rent and Non-Housing Components of CPI



Notes: This figure plots the Consumer Price Index, excluding shelter, energy, and food in pink. The line in black shows the rental price component of the CPI.

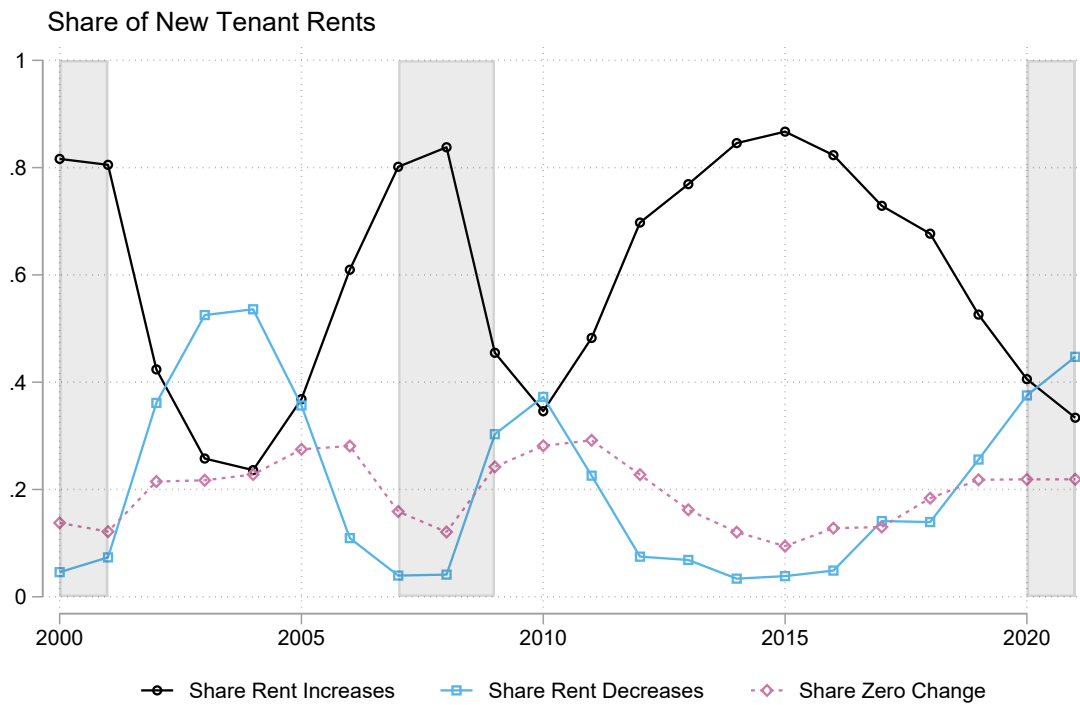
Figure 2: Evidence for State-Dependent Downward Nominal Rent Rigidity



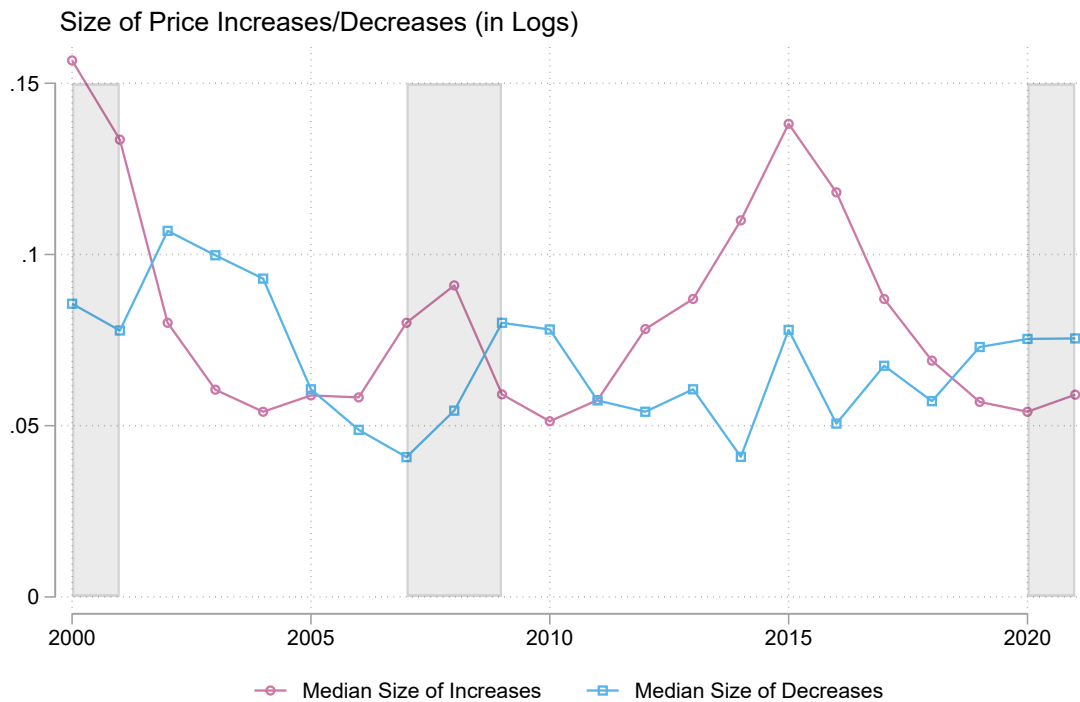
Notes: Panel (A) shows a histogram of the share of new leases with each nominal rent change, rounded to the nearest dollar. Panel (B) shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that have an absolute value of less than 20 log points. Panel (C) shows a binned scatter plot of a regression of an unchanged-rent indicator on log months since the previous lease. The regression includes neighborhood-year and unit fixed effects. Point estimates and standard errors clustered at the unit level are shown in parentheses. Panel (D) shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that have an absolute value of less than 20 log points. We split the sample into terciles of annual rent growth, based on an estimated repeat-rent index, and show the estimates separately by tercile of annual rent growth.

Figure 3: Frequency and Magnitude of Price Adjustments and the Business Cycle

A. Share of Rent Adjustments



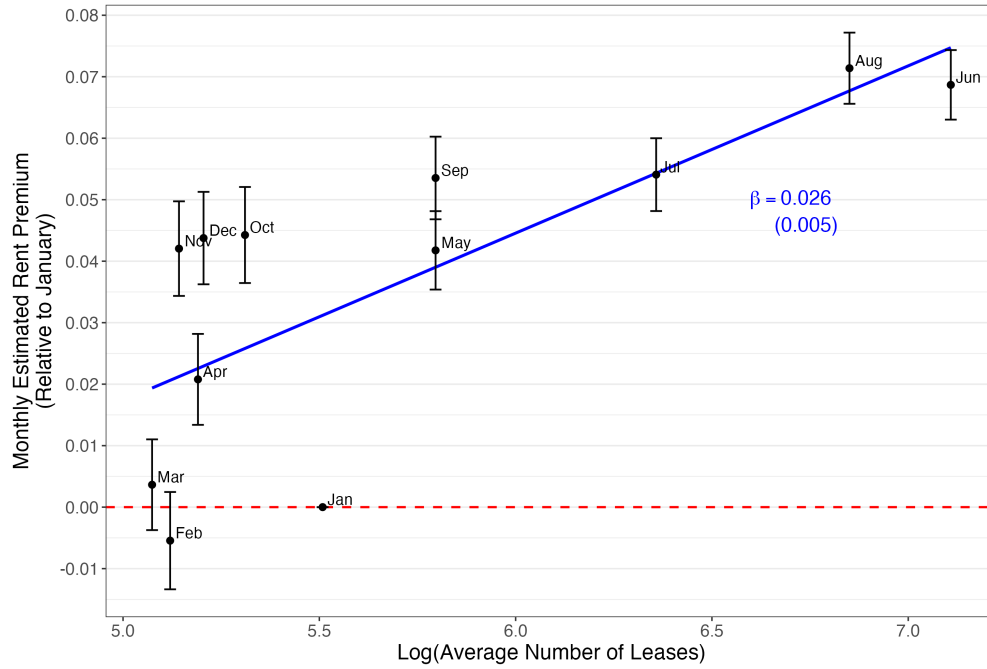
B. Size of Rent Adjustments



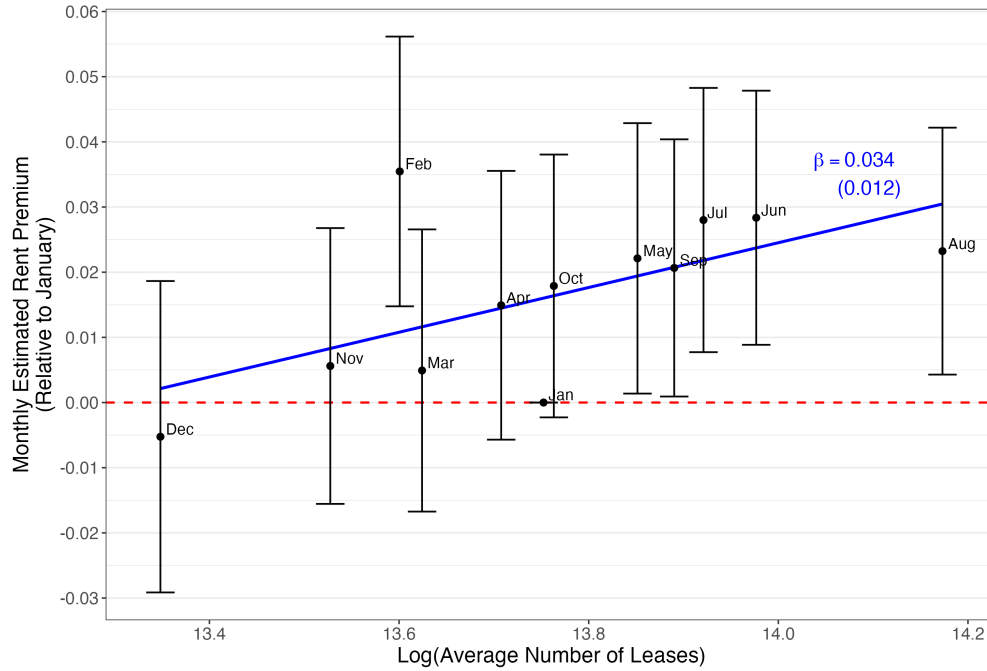
Notes: Panel (A) shows the share of new rental contracts that represent rent increases, rent decreases, or are unchanged in each year. Panel (B) shows the median size of rent increases and decreases separately. For both figures, we subset to units that turn over within two years.

Figure 4: Rental Price Premia and the Seasonal Cycle

A. Seasonal Price Premia in Berkeley Rental Market

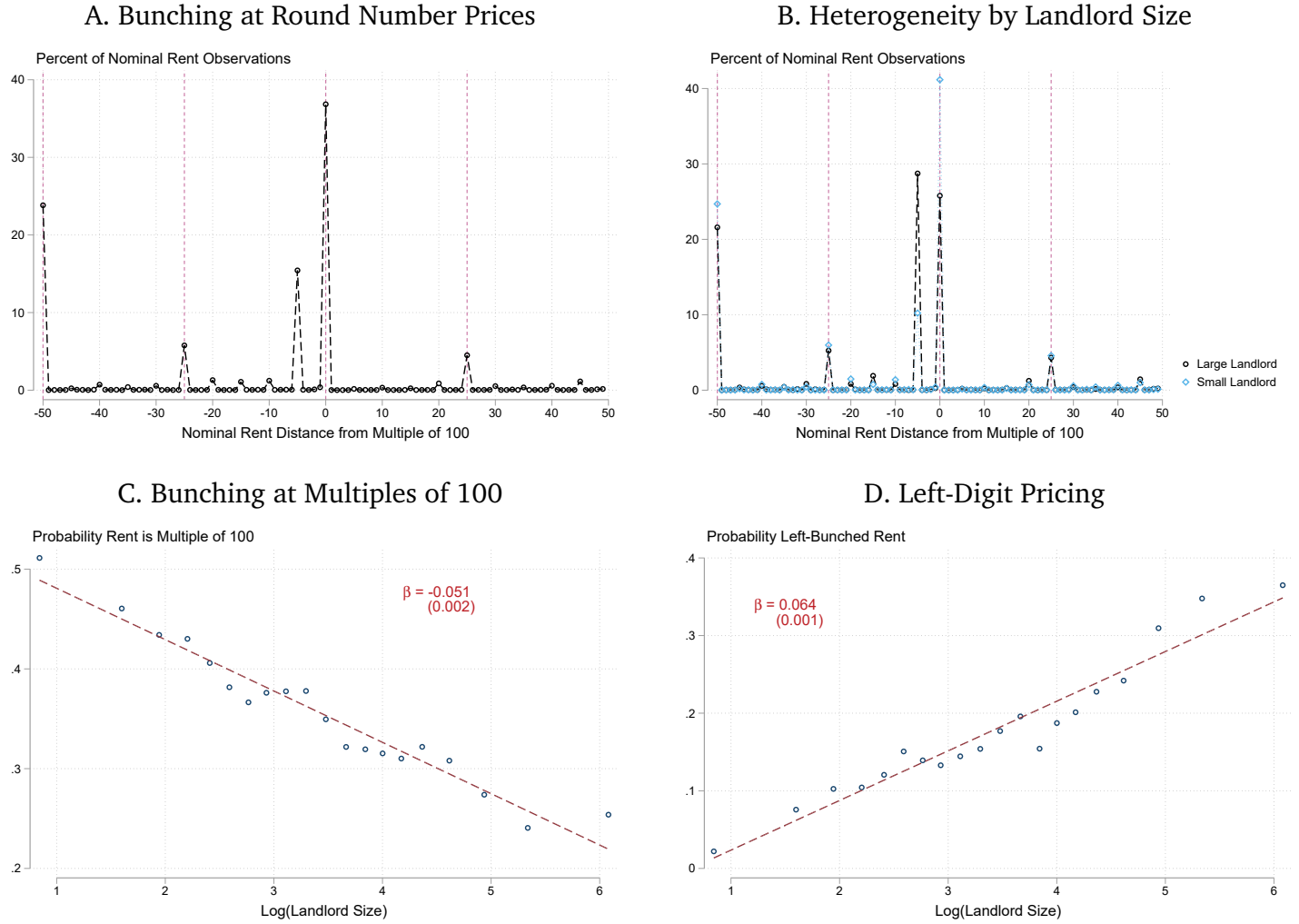


B. Seasonal Price Premia in American Rental Market



Notes: This figure shows estimated rental premia in both Berkeley and the broader American rental market. Panel (A) shows scatter plot of the estimated rent premia on the average number of leases per month (in logs) in Berkeley. Monthly rent premia are estimated from a regression of log rents on month fixed effects, controlling for neighborhood-year, month of last lease, number of months since last lease, and unit fixed effects. We only include pre-Covid years in these estimates, to avoid changes in the seasonal cycle caused by the pandemic. Estimated rent premia and standard errors are shown, as is the regression coefficient from a regression of the rent premia on the log(average number of leases). Point estimates and robust standard errors for that regression are shown. Panel (B) shows scatter plot of the estimated rent premia on the average number of leases per month (in logs) in the broader American market. Monthly rent premia are estimated from a regression of log rents on month fixed effects, controlling for city-by-year and unit fixed effects. Estimated rent premia and standard errors are shown, as is the regression coefficient of the rent premia on the log(average number of leases). Point estimates and robust standard errors for that regression are shown.

Figure 5: Evidence for Heterogeneity in Landlord Sophistication



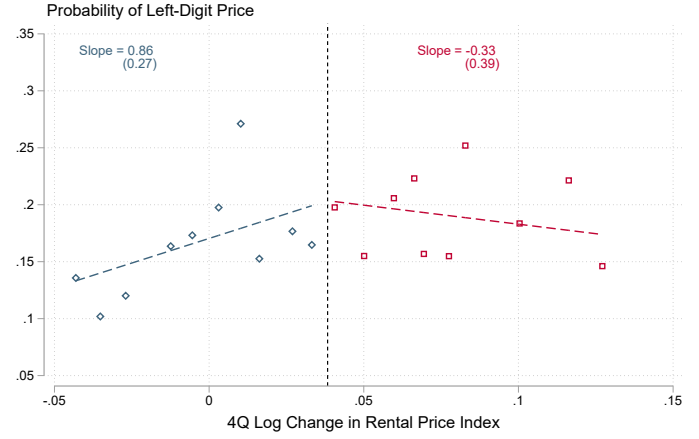
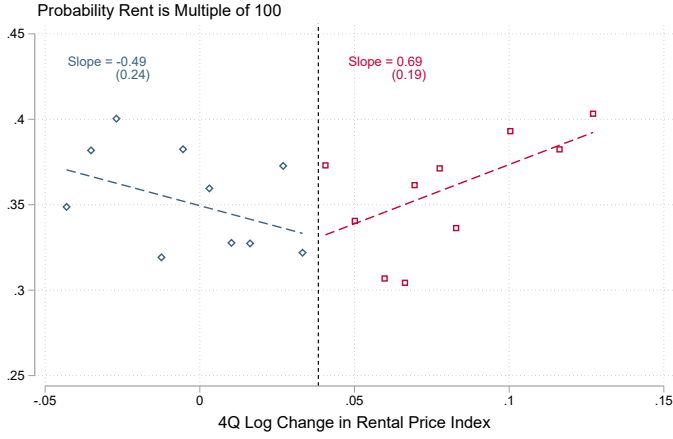
Notes: Panel (A) shows the percent of all observations in the dataset with numbers based on their distance from the nearest multiple of \$100. Panel (B) shows the same figures separately for large landlords (defined as landlords with 75 or more units in Berkeley) and small landlords (all other observations). Panels (C) and (D) show binned scatter plots of different rent characteristics on Log(Landlord Size). All regressions include month-of-lease-start by month-of-last-lease-start and census-tract-by-year fixed effects. In Panel (C), the outcome variable is the probability that the rent is an exact multiple of \$100. In Panel (D), the outcome variable is the probability of a left-bunched rent, defined as the probability that a rent observation is \$1 to \$9 less than a multiple of \$100.

Figure 6: Coarse Pricing Patterns Across The Business Cycle

Aggregate Effects

A. Bunching at Multiples of 100

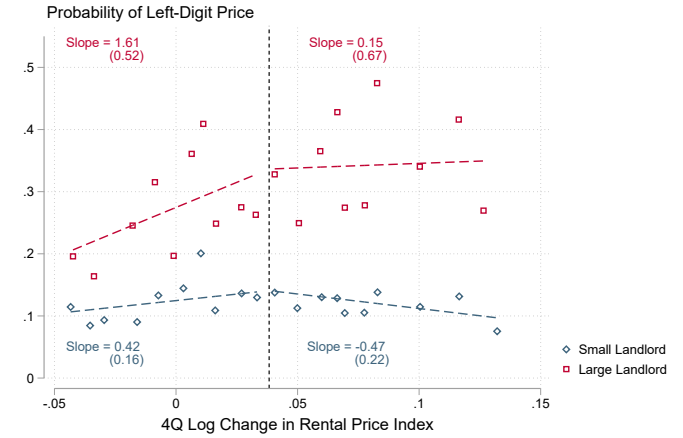
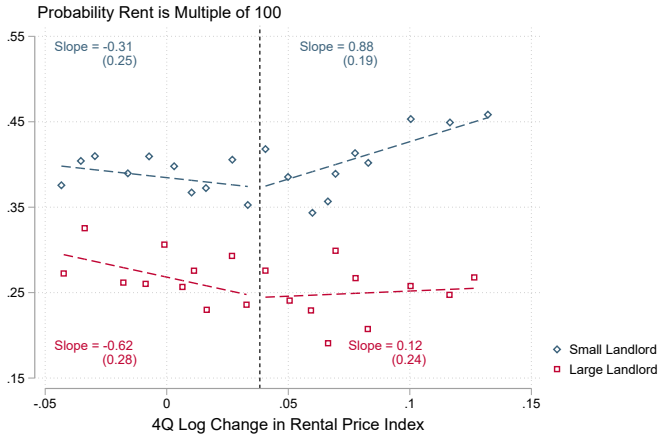
B. Left-Digit Pricing



Heterogeneity by Landlord Size

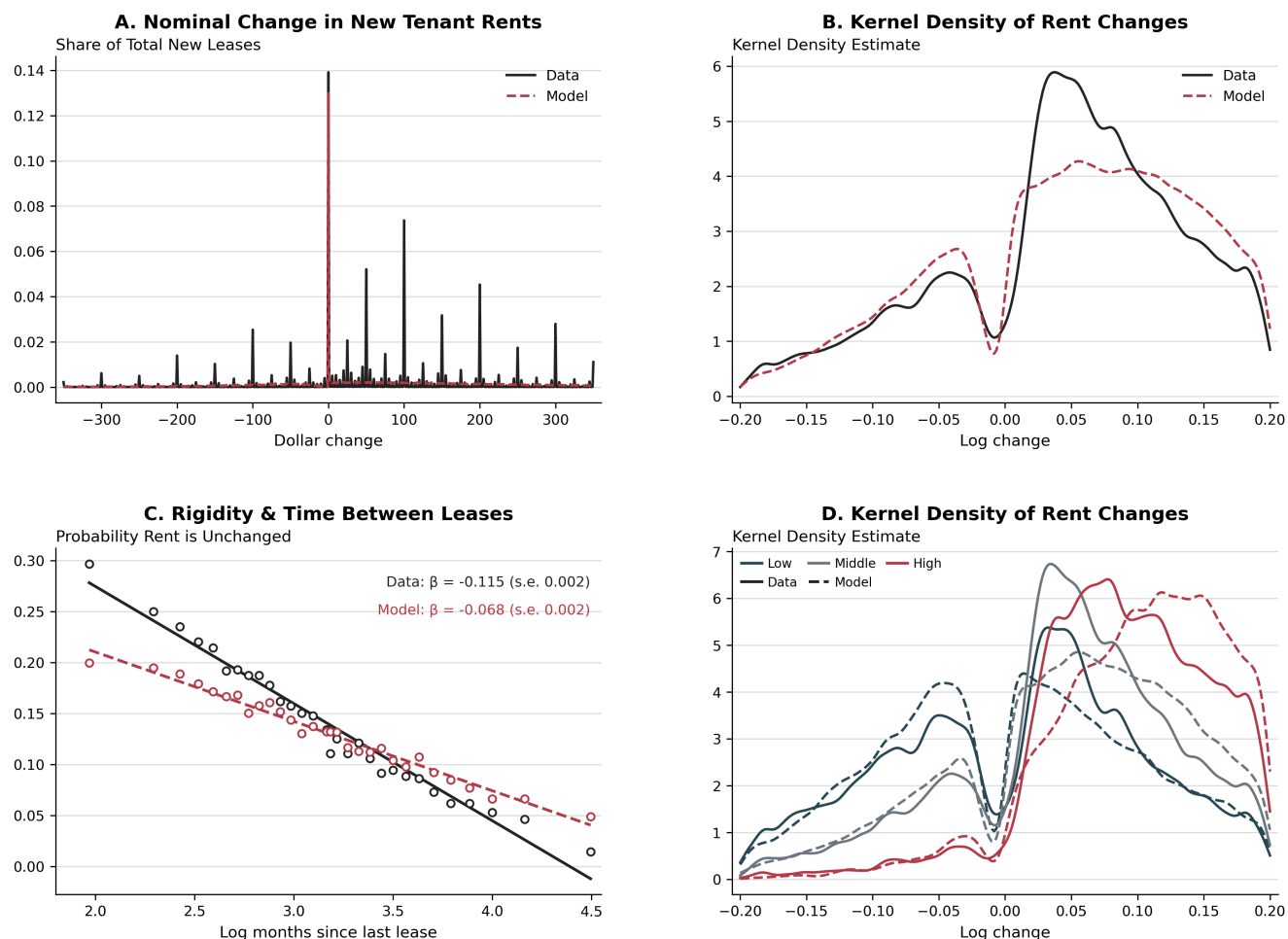
C. Bunching at Multiples of 100

D. Left-Digit Pricing



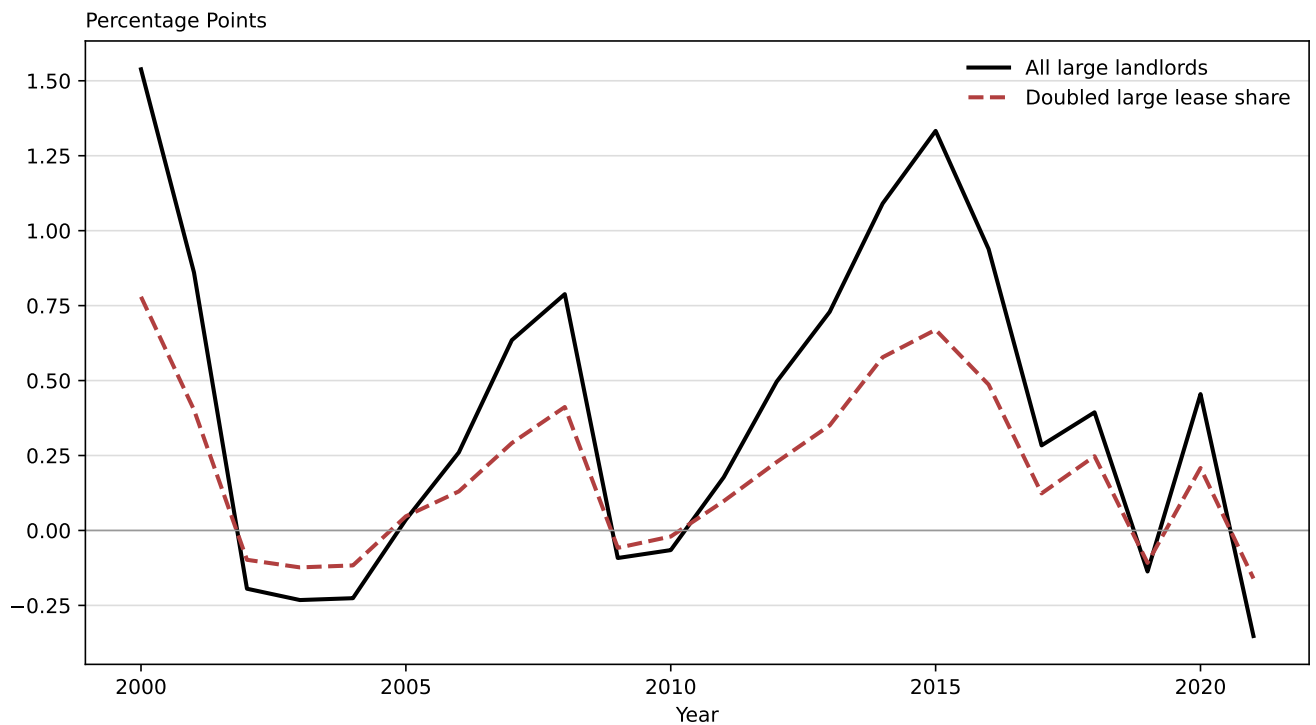
Notes: Panels (A) and (B) show binned scatter plots of the probability that a rent matches a particular characteristic on the annual change in the rental price index. We estimate separate slopes around a kink at median price growth in our sample. In Panel (A), the outcome variable is the probability the rent is a multiple of \$100. In Panel (B), the outcome variable is the probability of a left-digit price, defined as any price where the rent is \$1 to \$9 below a multiple of \$100. Outcome variables are residualized on the months since previous lease in logs. Panels (C) and (D) show the same figures, but separately for large and small landlords, defined based on whether a landlord owns 75 or more units in Berkeley. Outcome variables are residualized on the time between leases. Point estimates and standard errors clustered by date are shown.

Figure 7: Cross-Sectional Moments in the Model and Data



Notes: This figure reproduces plots from Figure 2, in both the actual data and in model-simulated data. Panel (A) shows a histogram of the share of new leases with each nominal rent change, rounded to the nearest dollar. Panel (B) shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that have an absolute value of less than 20 log points. Panel (C) shows a binned scatter plot of a regression of an unchanged-rent indicator on log months since the previous lease. The regression includes neighborhood-year and unit fixed effects. Point estimates and standard errors clustered at the unit level are shown in parentheses. Panel (D) shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that have an absolute value of less than 20 log points. We split the sample into terciles of annual rent growth, based on an estimated repeat-rent index, and show the estimates separately by tercile of annual rent growth.

Figure 8: The Effect of Large Landlords on Rents for New Tenants



Notes: This figure shows the effect of increasing the share of large landlords on new tenant rent inflation in our model. The line in black shows the effect of replacing all small landlords with large landlords, while the effect in red shows the effect of doubling the share of leases with large landlords. All effects are differences relative to the baseline new-tenant rent index from the model.

8 Tables

Table 1: Summary Statistics

Panel A: Berkeley Sample						
	N	Mean	Median	SD	P10	P90
Nominal Rent	92,370	1,692	1,495	1,142	850	2,800
Change in Rents	92,370	199	100	982	-100	605
Months Between Leases	92,370	34	24	31	11	70
Units at Address	92,370	19	12	23	2	44
Units per Landlord	92,370	88	28	140	4	287
Number of Bedrooms	92,370	1	1	1	0	2

Panel B: U.S. Sample						
	N	Mean	Median	SD	P10	P90
Nominal Rent	55,451	617	550	336	280	1,000
Change in Rents	55,451	73	45	224	-75	280
Units in Building	55,451	17	6	41	1	40
Number of Bedrooms	55,451	2	2	1	1	3

Notes: This table shows summary statistics for both our main samples. Panel (A) shows summary statistics for the Berkeley sample, while Panel (B) shows summary statistics for the broader U.S. sample.

Table 2: Aggregate Dynamics of Price Adjustment

Panel A: Frequency of Price Adjustment

	Tot. Leases	Increases	Decreases	Unchanged
Annual Log Rent Growth	-0.21 (0.47)	4.41*** (1.24)	-18.5*** (1.53)	-5.48*** (1.35)
<i>N</i>	22	22	22	22
<i>R</i> ²	0.0085	0.45	0.84	0.33

Panel B: Median Size of Price Adjustments

	≤ 2 Yrs. Since Last Lease		All Leases	
	Increase	Decrease	Increase	Decrease
Annual Log Rent Growth	0.54*** (0.088)	-0.19** (0.091)	0.56** (0.20)	-0.15** (0.070)
<i>N</i>	22	22	22	22
<i>R</i> ²	0.73	0.27	0.28	0.27

Notes: This table shows the relationship between changes in rental prices and various measures of price adjustment. In all regressions, the independent variable is the change in the rental price index (in logs). Panel A reports correlations with the frequency of price adjustments across different categories. All frequency variables are in logs. Panel B shows the relationship with the median size of price increases and decreases. All size variables are in log changes. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Dynamics of Rent Adjustment, Controlling for Unit Fixed Effects

Panel A: Berkeley Sample

	Share of Rent Changes			Size of Rent Changes		
	Increases	Decreases	Unchanged	Increases	Decreases	Combined
4Q Log Rent Index	3.244*** (0.229)	-2.594*** (0.213)	-0.650*** (0.075)	0.711*** (0.135)	-0.348*** (0.089)	1.291*** (0.130)
<i>N</i>	87452	87452	87452	59199	10337	87452
<i>R</i> ²	0.306	0.265	0.258	0.487	0.429	0.408
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.20	0.11	0.12

Panel B: U.S. Sample

	Share of Rent Changes			Size of Rent Changes		
	Increases	Decreases	Unchanged	Increases	Decreases	Combined
4Q Log Rent Index	3.342*** (0.603)	-2.804*** (0.507)	-0.539** (0.232)	0.905*** (0.318)	-0.732 (1.360)	2.076*** (0.528)
<i>N</i>	50356	50356	50356	33587	6265	50356
<i>R</i> ²	0.208	0.219	0.274	0.431	0.522	0.146
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.70	0.22	0.08	0.27	0.37	0.12

Notes: This table shows the relationship between changes in rental price indexes and the size and frequency of price adjustment at the unit level. In the first three columns, the dependent variables are indicators for whether rents increased, decreased, or were unchanged respectively. In the next three columns, the dependent variables are the size of price increases, price decreases, and all price changes, in logs. All regressions include individual unit fixed effects. Panel (A) shows results for the Berkeley sample, while Panel (B) shows results for the U.S. sample. The regressions in Panel (A) also include controls for the month in which the previous lease started. Standard errors clustered by date in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Pricing Response to the Seasonal Cycle, Controlling for Unit Fixed Effects

Panel A: Berkeley Sample

	Share of Rent Changes			Size of Rent Changes		Total
	Increases	Decreases	Unchanged	Increases	Decreases	Log(Rent)
Avg. Num. Leases	0.028*** (0.004) [0.000]	-0.025*** (0.003) [0.000]	-0.004 (0.002) [0.160]	-0.001 (0.004) [0.747]	-0.010** (0.003) [0.030]	0.024*** (0.004) [0.000]
<i>N</i>	82828	82828	82828	57822	8018	82828
<i>R</i> ²	0.374	0.325	0.289	0.555	0.589	0.938
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Nbhood-Yr	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.71	0.16	0.13	0.20	0.11	7.28

Panel B: U.S. Sample

	Share of Rent Changes			Size of Rent Changes		Total
	Increases	Decreases	Unchanged	Increases	Decreases	Log(Rent)
Avg. Num. Leases	0.057*** (0.012) [0.010]	-0.034** (0.011) [0.039]	-0.024** (0.005) [0.018]	0.004 (0.008) [0.604]	0.015 (0.069) [0.810]	0.039* (0.014) [0.068]
<i>N</i>	45068	45068	45068	29603	4743	45068
<i>R</i> ²	0.307	0.308	0.335	0.495	0.678	0.703
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
SMSA-Yr	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.70	0.22	0.08	0.26	0.37	6.30

Notes: This table shows how rental pricing varies across the seasonal cycle. The independent variable is the average number of leases in each month across the sample periods. Columns (1) - (6) show the effect of the seasonal cycle on rental prices. Panel (A) shows the effect in the Berkeley sample. Regressions represented by Columns (1) - (6) in the Berkeley sample include fixed effects for the neighborhood-by-year, the number of months between leases, the month the previous lease started, and the individual unit. Panel (B) shows the effects in the broader U.S. sample. Regressions represented by Columns (1) - (6) in the U.S. sample include city-year and unit fixed effects, including a residual category for units that are not identified with a particular city for disclosure reasons. Standard errors clustered by month in parentheses. P-values computed via the wild clustered bootstrap to account for the small number of clusters in brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Causal Effect of Landlord Size on Coarse Pricing Using Sold Units

	Rent is...		
	Bunch at 50	Bunch at 100	Left-Digit Pricing
Landlord Size	-0.037*** (0.004)	-0.022*** (0.003)	0.041*** (0.003)
<i>N</i>	56295	56295	56295
<i>R</i> ²	0.407	0.393	0.429
Nbhood-Qtr FE	Yes	Yes	Yes
Unit FE	Yes	Yes	Yes
Dep. Mean	0.58	0.34	0.20

Notes: This table shows the causal effect of landlord size on bunching. The regressions are run in a sample of units that are sold during the sample period, and all regressions include unit fixed effects. The independent variable is the maximum number of units held at any time by the landlord, so all variation comes from the sale between different landlords. All regressions include month-of-lease-start by month-of-last-lease-start, months-since-last-lease, and neighborhood-by-quarter fixed effects. Standard errors clustered by unit in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Inflation Volatility and Heuristic Pricing, Controlling for Unit Fixed Effects

Panel A: Overall Effect			
	Bunch at 50	Bunch at 100	Left-Digit Price
Slope Above	0.792*** (0.227)	0.735*** (0.187)	-0.308 (0.335)
Slope Below	-0.399* (0.213)	-0.368* (0.204)	0.521** (0.223)
<i>N</i>	87452	87452	87452
<i>R</i> ²	0.385	0.355	0.396
Unit FE	Yes	Yes	Yes
Dep. Mean	0.59	0.35	0.18

Panel B: Split by Landlord Size			
<i>Panel (A): Small Landlords (< 75 Units)</i>			
	Bunch at 50	Bunch at 100	Left-Digit Price
Slope Above	0.998*** (0.181)	0.903*** (0.189)	-0.406** (0.174)
Slope Below	-0.209 (0.225)	-0.245 (0.216)	0.232* (0.135)
<i>N</i>	59572	59572	59572
<i>R</i> ²	0.416	0.384	0.399
Unit FE	Yes	Yes	Yes
Dep. Mean	0.65	0.40	0.12
<i>Panel (B): Large Landlords (≥ 75 Units)</i>			
	Bunch at 50	Bunch at 100	Left-Digit Price
Slope Above	0.133 (0.327)	0.184 (0.227)	0.249 (0.514)
Slope Below	-0.491* (0.291)	-0.438* (0.246)	0.808** (0.379)
<i>N</i>	26528	26528	26528
<i>R</i> ²	0.371	0.315	0.423
Unit FE	Yes	Yes	Yes
Dep. Mean	0.48	0.26	0.30

Notes: This table shows the relationship between deviations of rental price inflation from its usual trend and measures of heuristic pricing. The regression estimates separate slopes for the dependent variable, corresponding to the distance from the median price change over the sample period, with a common intercept. In Column (1), the dependent variable is whether a rent is a multiple of 50. In Column (2), the dependent variable is whether a rent is a multiple of 100. In Column (3), the dependent variable is whether or not a rent shows evidence of "left-digit pricing", defined as being between \$1 and \$9 less than a multiple of 100. Panel (A) shows the pooled Berkeley sample, while Panel (B) shows the effect separately for large and small landlords. All estimates include unit fixed effects and controls for the number of months between lease dates. Standard errors clustered by date in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Heterogeneity in Pricing Behavior Over the Business Cycle

Panel A: Pooled Effects					
	Rent Change is...				
	Increase	Decrease	Unchanged	Big Increase	Log Change
Landlord Size	-0.002 (0.002)	0.006*** (0.002)	-0.004*** (0.002)	0.009*** (0.002)	0.002** (0.001)
<i>N</i>	89745	89745	89745	89745	89745
<i>R</i> ²	0.263	0.213	0.127	0.379	0.431
Nbhood-Qtr FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.25	0.12

Panel B: Business Cycle Effects					
	Rent Change is...				
	Increase	Decrease	Unchanged	Big Increase	Log Change
Landlord Size	-0.012*** (0.003)	0.019*** (0.004)	-0.006*** (0.002)	0.002 (0.003)	-0.002 (0.002)
x Mod. Growth	0.010* (0.005)	-0.012** (0.005)	0.002 (0.003)	0.005 (0.004)	0.004* (0.002)
x High Growth	0.021*** (0.005)	-0.025*** (0.005)	0.004 (0.004)	0.014** (0.005)	0.007*** (0.002)
<i>N</i>	89745	89745	89745	89745	89745
<i>R</i> ²	0.263	0.214	0.127	0.380	0.431
Nbhood-Qtr FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.25	0.12

Notes: This table shows how landlords of different sizes adjust prices over the business cycle. Panel (A) shows the pooled (or average) effect of large landlords over the entire sample period. Panel (B) shows the effect over the business cycle. The coefficients of interest in Panel (B) are interaction terms between landlord size (in logs) and terciles of rent price growth in Berkeley, California. The omitted category is the lowest tercile of the rent growth distribution. Both sets of estimates include neighborhood-by-quarter fixed effects, as well as fixed effects for the month the previous lease started by month the current lease started (to account for seasonal effects), as well as the number of months between leases. Standard errors double clustered by unit and date in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Causal Effect of Landlord Heterogeneity on Cyclical Sensitivity

	Rent Change is...				
	Increase	Decrease	Unchanged	Big Increase	Log Change
Landlord Size	-0.006 (0.004)	0.009** (0.004)	-0.003 (0.003)	-0.004 (0.005)	0.003 (0.003)
x Mod. Growth	0.005 (0.006)	-0.004 (0.005)	-0.000 (0.004)	0.001 (0.005)	-0.000 (0.002)
x High Growth	0.019*** (0.005)	-0.018*** (0.004)	-0.000 (0.004)	0.012** (0.006)	0.004* (0.002)
<i>N</i>	56295	56295	56295	56295	56295
<i>R</i> ²	0.410	0.367	0.295	0.487	0.517
Nbhood-Qtr FE	Yes	Yes	Yes	Yes	Yes
Unit FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.12	0.25	0.13

Notes: This table shows the causal effect of landlord heterogeneity on sensitivity to the business cycle. The regressions are run in a sample of units that are sold during the sample period, and all regressions include unit fixed effects. The independent variable is the maximum number of units held at any time by the landlord, so all variation comes from the sale between different landlords. These variables are interacted with indicators for different phases of the business cycle. All estimates include neighborhood-by-quarter fixed effects, as well as fixed effects for the month the previous lease started by month the current lease started (to account for seasonal effects), as well as the number of months between leases. Standard errors double clustered by unit and date in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Model Parameters and Targeted Moments

Panel A: Parameter Estimates		
Parameter		Value
Residual SD, small	$\sigma_{z,S}$	0.0724
Residual SD, large	$\sigma_{z,L}$	0.0586
Index projection, small	θ_S	0.9592
Index projection, large	θ_L	1.1106
Reopt. cost, small	κ_S^r	0.0109
Reopt. cost, large	κ_L^r	0.0144
Cost prob., small	λ_S^r	0.1031
Cost prob., large	λ_L^r	0.2311
Cut cost, small	κ_S^c	0.00115
Cut cost, large	κ_L^c	0.00001

Panel B: Fit of Targeted Moments		
Targeted Moment	Model	Data
<i>Cross-Sectional</i>		
SD log change	0.170	0.210
No change	0.126	0.139
Rent cut	0.184	0.185
Small cut (\$50 or less)	0.040	0.046
Small raise (\$50 or less)	0.099	0.122
Mean abs. log raise	0.173	0.183
<i>Lease-Age Gaps</i>		
0–12 months	0.180	0.250
13–24 months	0.147	0.156
25–60 months	0.097	0.081
61+ months	0.042	0.027
<i>Landlord Size Gaps</i>		
SD log change	-0.025	-0.025
No change	0.013	0.018
Rent cut	0.003	0.005
Mean abs. log raise	-0.013	-0.021
Large IQR log change, 25–60 minus 0–12	0.130	0.141
Small IQR log change, 25–60 minus 0–12	0.126	0.122

Notes: This table shows the fit of the quantitative model. Panel (A) shows the estimated parameters from the model, while Panel (B) shows the targeted moments in both the model and the data.

Table 10: Dynamics of Price Adjustment by Landlord Size: Data and Model

Panel A: Data					
	Rent Change is...				
	Increase	Decrease	Unchanged	Big Increase	Log Change
Large Landlord	-0.030** (0.013)	0.046*** (0.013)	-0.016** (0.006)	0.004 (0.009)	-0.005 (0.006)
x Mod. Growth	0.015 (0.016)	-0.037** (0.015)	0.022** (0.010)	0.006 (0.013)	0.008 (0.008)
x High Growth	0.037** (0.018)	-0.061*** (0.015)	0.024** (0.012)	0.037** (0.016)	0.018** (0.008)
N	85075	85075	85075	85075	85075
R ²	0.262	0.223	0.126	0.336	0.389
Nbhood-QTR FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.68	0.18	0.14	0.21	0.10

Panel B: Model					
	Rent Change is...				
	Increase	Decrease	Unchanged	Big Increase	Log Change
Large Landlord	-0.048*** (0.011)	0.044*** (0.010)	0.004 (0.004)	0.001 (0.004)	0.004 (0.003)
x Mod. Growth	0.048*** (0.013)	-0.047*** (0.013)	-0.000 (0.006)	0.016** (0.007)	0.009** (0.004)
x High Growth	0.064*** (0.012)	-0.067*** (0.011)	0.004 (0.006)	0.042*** (0.010)	0.017*** (0.003)
N	85075	85075	85075	85075	85075
R ²	0.213	0.173	0.071	0.373	0.475
Nbhood-QTR FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.21	0.10

Notes: This table shows regressions in both the data and the model-simulated data estimating how rent changes respond to the business cycle. Panel (A) shows the estimates from the data, while Panel (B) shows the estimates from the model-generated data. Regressions include neighborhood-by-quarter, previous-lease-month-by-current-lease-month, and months-since-last-lease fixed effects. Standard errors double clustered by date and unit in parentheses.

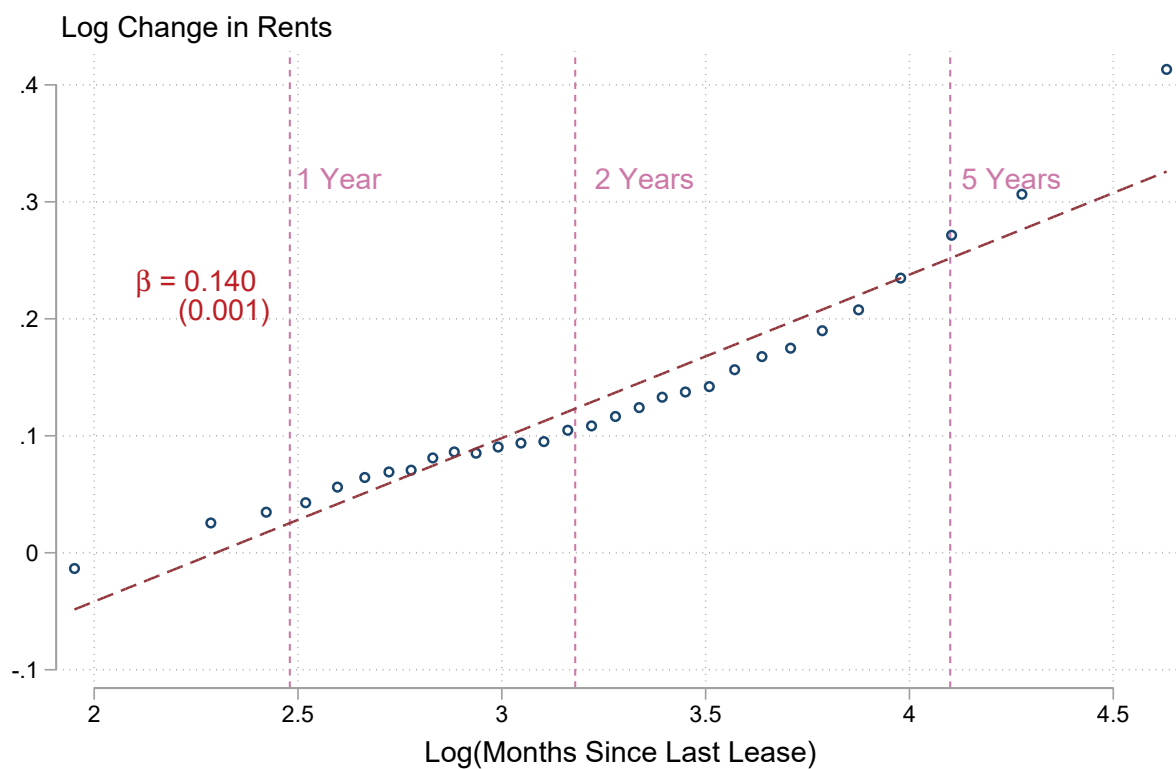
A Appendix Figures

Figure A.1: Prices in the Berkeley Rental Market



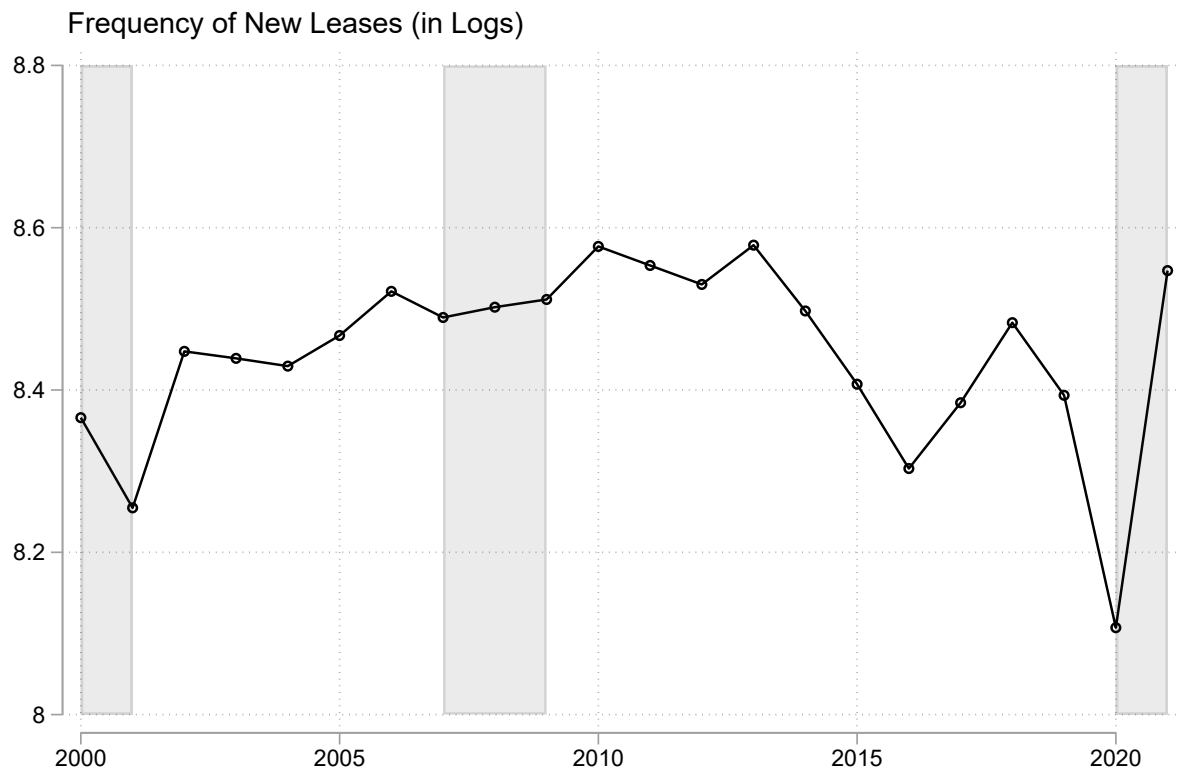
Notes: This figure plots the new-tenant rent price index in Berkeley, CA, extracted from a regression of new tenant rents on year and unit fixed effects.

Figure A.2: Reset Prices Increase with the Time Between Leases



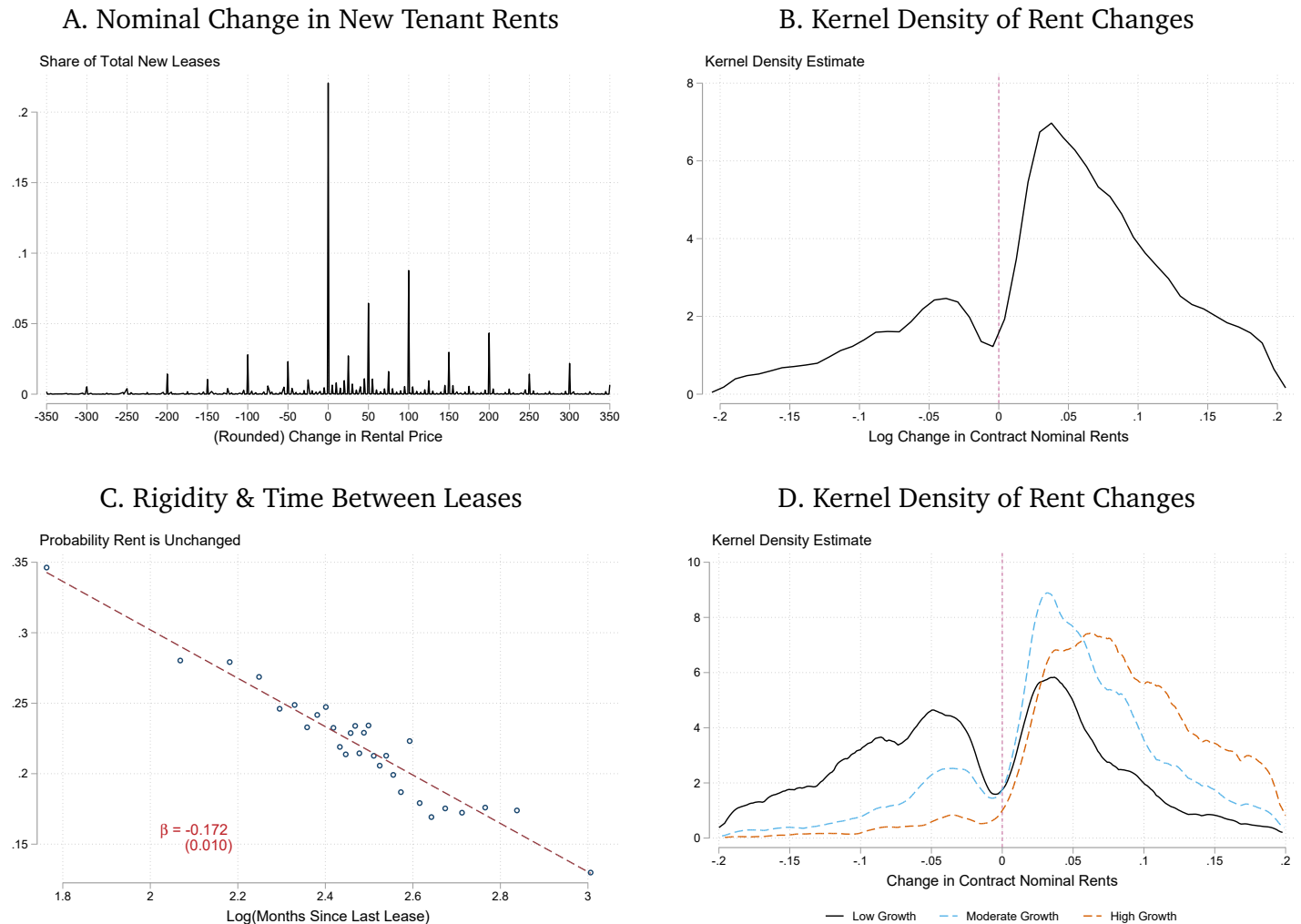
Notes: This figure shows binned scatter plots of the change in log rental prices on the number of months since the last lease (in logs). The regression includes census-tract-by-year and unit fixed effects. Point estimates and standard errors clustered at the unit level are shown.

Figure A.3: Aggregate Mobility Does Not Vary Over Time



Notes: This figure shows the time series of annual new leases in Berkeley, CA.

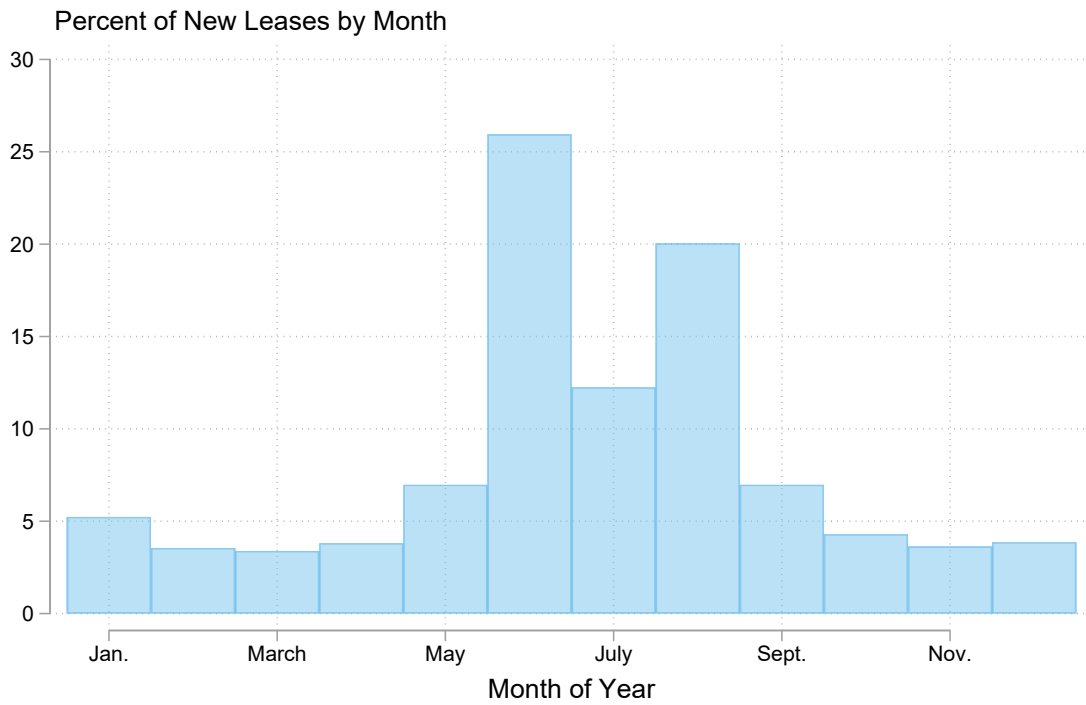
Figure A.4: State-Dependent Downward Nominal Rent Rigidity for Leases without Rent Adjustments



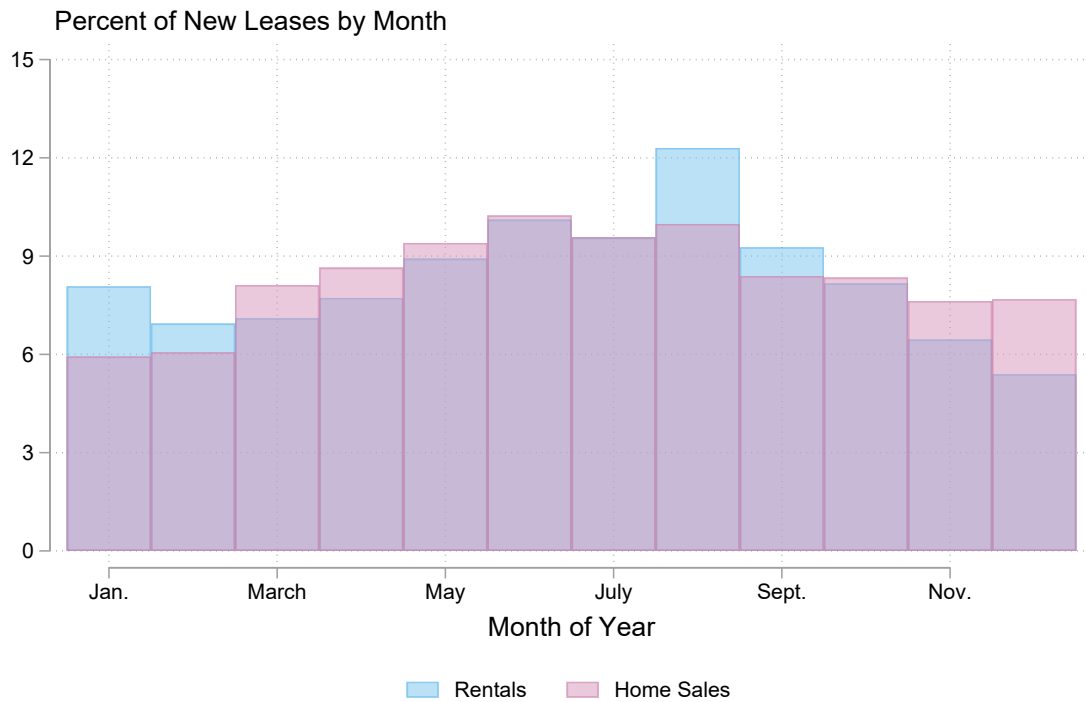
Notes: This figure reproduces Figure 2 for a sample of lease changes that are not eligible for any indexed rent changes. Panel (A) shows a histogram of the share of new leases with each nominal rent change, rounded to the nearest dollar. Panel (B) shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that have an absolute value of less than 20 log points. Panel (C) shows a binned scatter plot of a regression of an unchanged-rent indicator on log months since the previous lease. The regression includes neighborhood-year and unit fixed effects. Point estimates and standard errors clustered at the unit level are shown in parentheses. Panel (D) shows the kernel density estimate of log rent changes after excluding observations with zero rent change. For readability, we only include rent changes that have an absolute value of less than 20 log points. We split the sample into terciles of annual rent growth, based on an estimated repeat-rent index, and show the estimates separately by tercile of annual rent growth.

Figure A.5: Number of New Leases by Month

A. Berkeley Sample

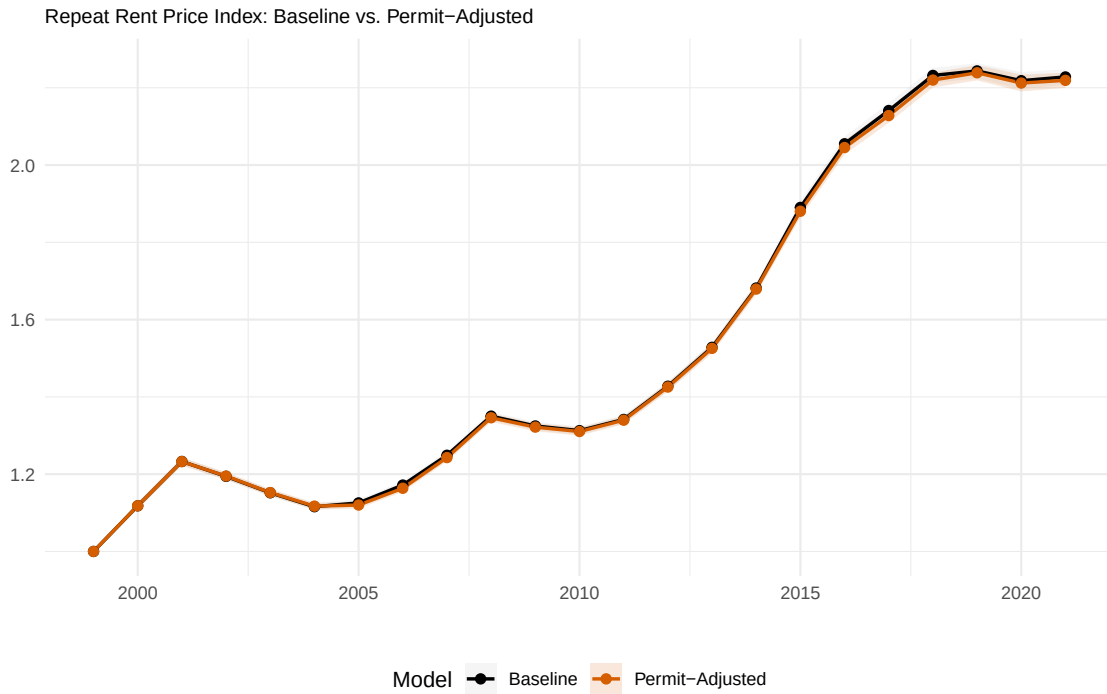


B. U.S. Sample



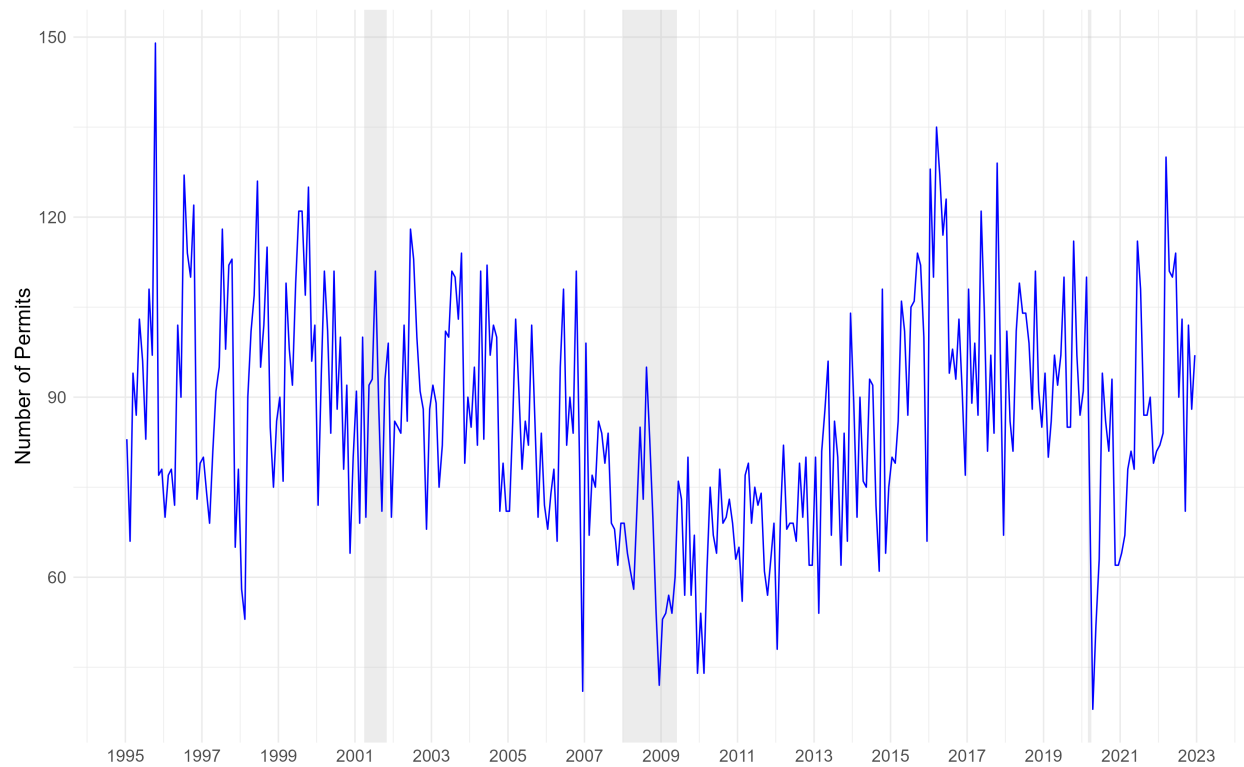
Notes: Panel (A) shows the percent of new leases by month in the Berkeley sample. Panel (B) shows the same thing but for the broader U.S. sample. Panel (B) also shows seasonality in the home purchase market for the entire U.S., taken from sales data from the National Association of Realtors in Ngai and Tenreyro (2014). For the Berkeley sample, we end the sample in 2020 to avoid any unusual changes due to the pandemic.

Figure A.6: Permit-Adjusted Prices in the Berkeley Rental Market



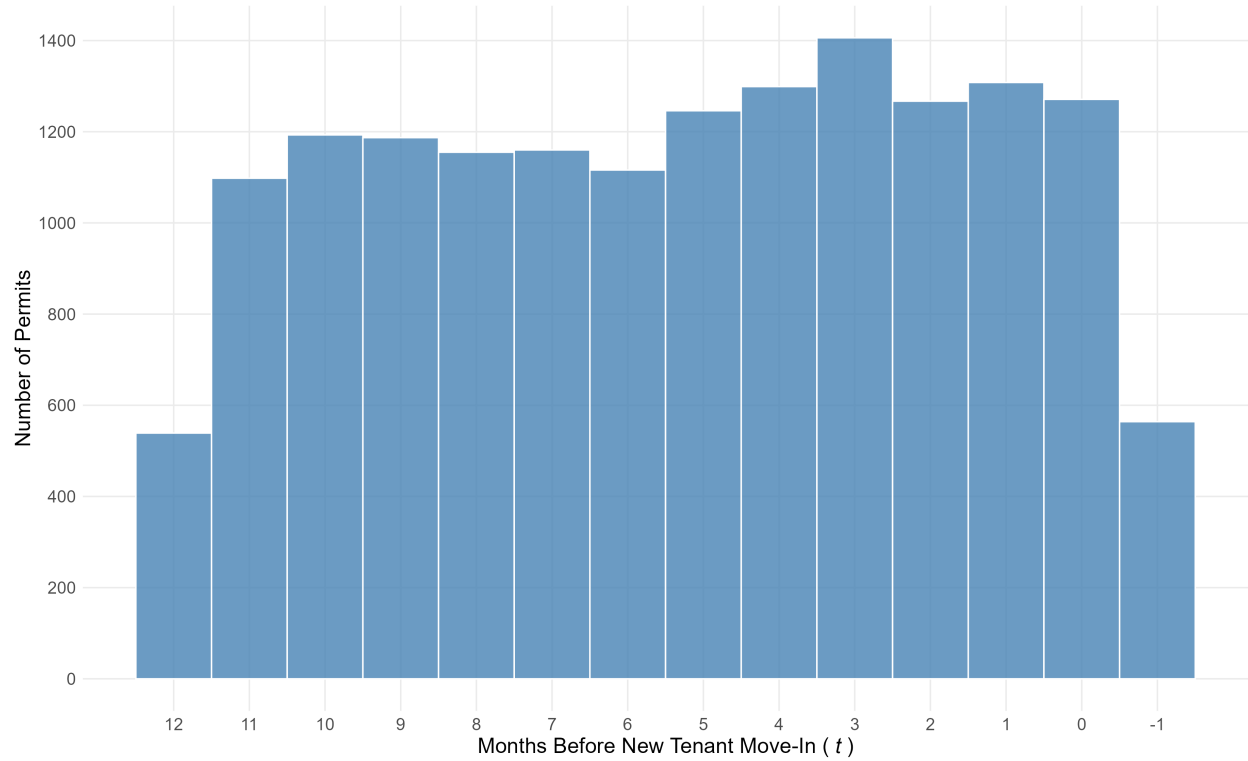
Notes: This figure plots the permit-adjusted new-tenant rent price index in Berkeley, CA, extracted from a regression of new tenant rents on year, unit fixed effects, and permit scores.

Figure A.7: Permits Over Time in Berkeley



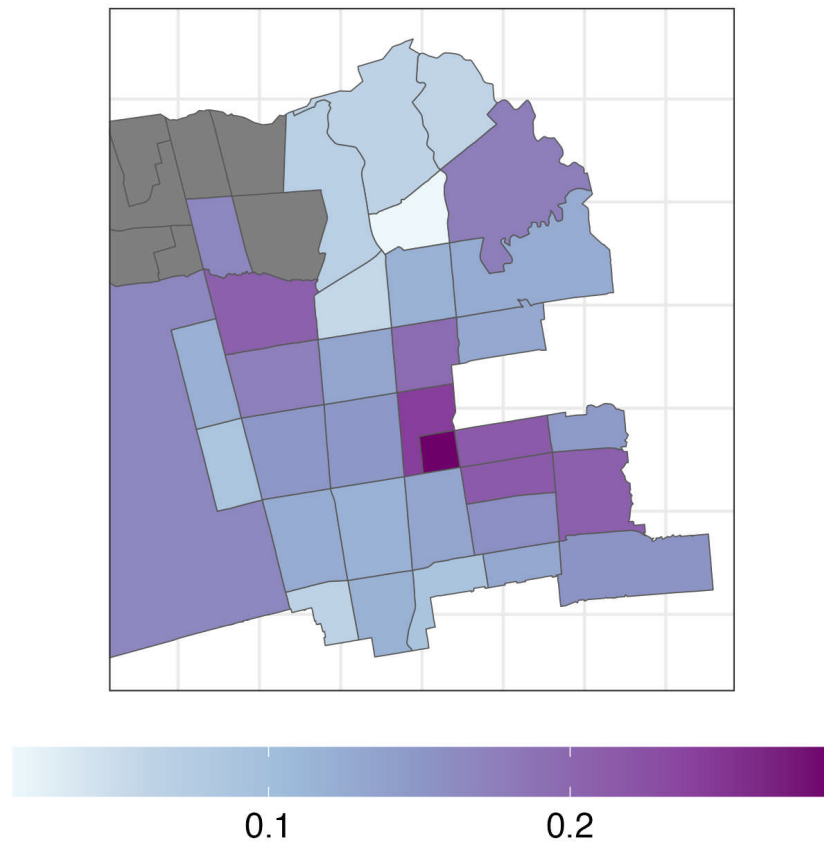
Notes: This figure shows the number of building permits filed over time for Berkeley rental properties in our sample. Recessions are represented by gray shaded bars.

Figure A.8: Permits Over Lease Length in Berkeley



Notes: This figure shows the number of building permits filed over the 12 months of a lease prior to a new tenant's move-in. Permit activity is relatively stable over this period, with a slight uptick in the last five months of a tenancy ($t \in [5, 0]$) and a drop off in activity during the first month of a new tenancy ($t = -1$).

Figure A.9: Map of Left-Digit Pricing in Berkeley, California



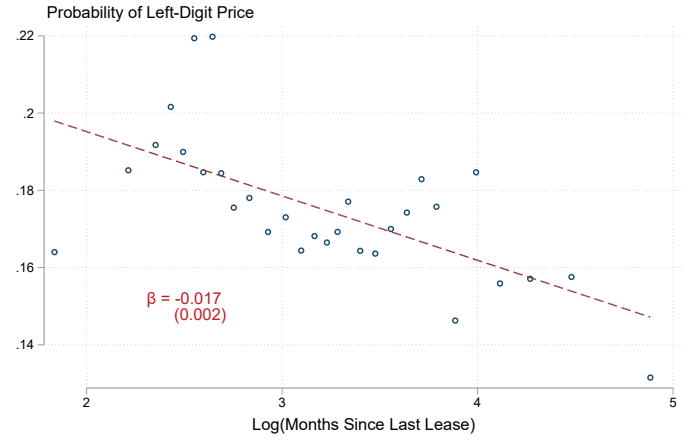
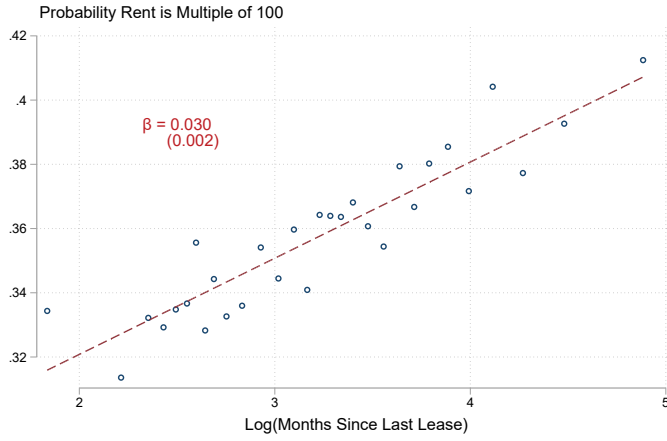
Notes: This figure shows a census tract level map with the average level of left-digit pricing in each census tract in Berkeley.

Figure A.10: Coarse Pricing Patterns Across Lease Contract Gaps

Aggregate Effects

A. Bunching at Multiples of 100

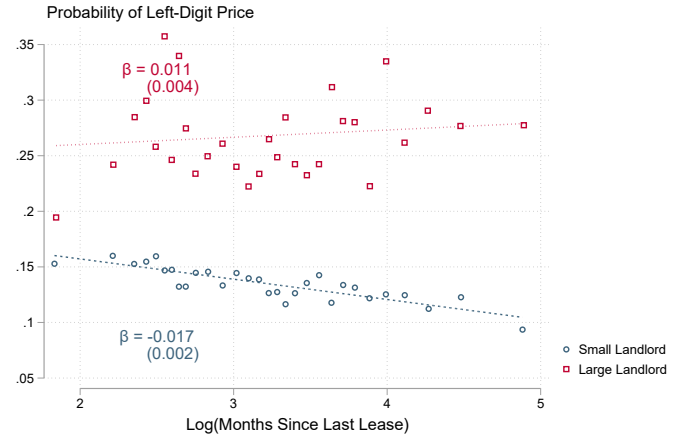
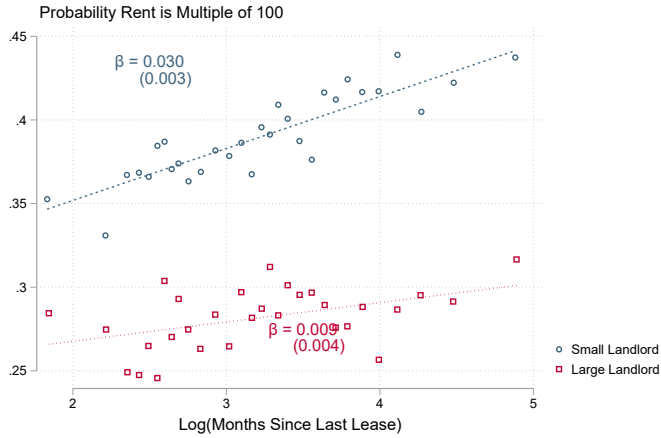
B. Left-Digit Pricing



Heterogeneity by Landlord Size

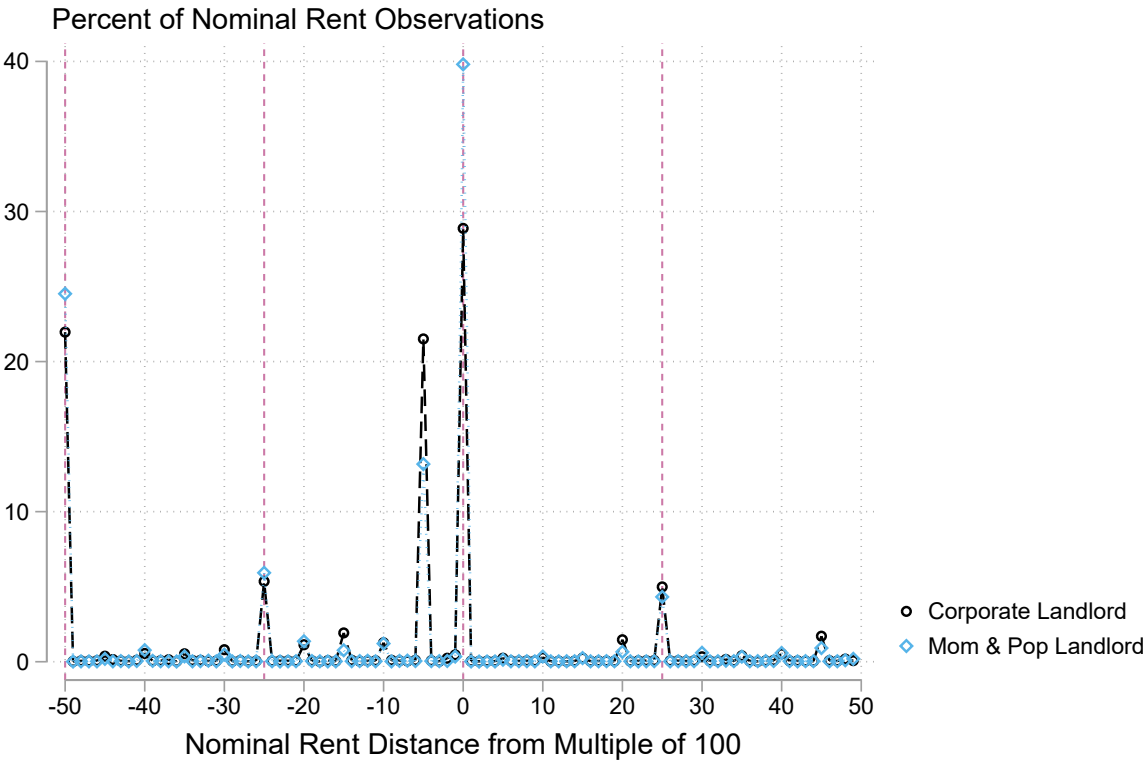
C. Bunching at Multiples of 100

D. Left-Digit Pricing



Notes: Panels (A) and (B) show binned scatter plots of the probability that a rent matches a particular characteristic on the time since the previous lease. In Panel (A), the outcome variable is the probability the rent is a multiple of \$100. In Panel (B), the outcome variable is the probability of a left-digit price, defined as any price where the rent is \$1 to \$9 below a multiple of \$100. Panels (C) and (D) show the same graphs, but separately for large (≥ 75 units in Berkeley) and small landlords. Outcome variables are residualized on census-tract-by-year fixed effects. Point estimates and standard errors clustered by unit are shown.

Figure A.11: Bunching for Corporate and Individual Landlords



Notes: This figure shows bunching in the rent distribution separately for corporate and mom & pop landlords.

B Appendix Tables

Table B.1: Aggregate Dynamics of Permit Timing

	Tot. Permits	Log Tot. Permits
Annual Log Rent Growth	408.25 (655.07)	0.44 (0.68)
Observations	22	22
R ²	0.02	0.02

Notes: This table shows the relationship between changes in rental prices and building permit activity. In all regressions, the independent variable is the change in the rental price index (in logs). Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Heterogeneity in Pricing Behavior Over the Business Cycle

Panel A: Pooled Effects					
	Rent Change is...				
	Increase	Decrease	Unchanged	Big Increase	Log Change
Corp. Landlord	0.003 (0.005)	0.001 (0.004)	-0.003 (0.004)	0.017*** (0.005)	0.011*** (0.003)
<i>N</i>	89745	89745	89745	89745	89745
<i>R</i> ²	0.263	0.213	0.126	0.379	0.431
Nbhood-Qtr FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.25	0.12

Panel B: Business Cycle Effects					
	Rent Change is...				
	Increase	Decrease	Unchanged	Big Increase	Log Change
Corp. Landlord	0.011 (0.010)	0.008 (0.010)	-0.018*** (0.006)	0.016*** (0.006)	0.009** (0.005)
x Mod. Growth	-0.013 (0.014)	-0.008 (0.013)	0.021** (0.010)	0.002 (0.010)	0.001 (0.006)
x High Growth	-0.009 (0.011)	-0.013 (0.011)	0.022*** (0.007)	0.002 (0.012)	0.005 (0.008)
<i>N</i>	89745	89745	89745	89745	89745
<i>R</i> ²	0.263	0.213	0.127	0.379	0.431
Nbhood-Qtr FE	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.25	0.12

Notes: This table shows how corporate vs. non-corporate landlords adjust prices over the business cycle. The coefficients of interest are interaction terms between an indicator for corporate landlords and terciles of rent price growth in Berkeley, California. The omitted category is the lowest tercile of the rent growth distribution. The regressions include neighborhood-by-quarter fixed effects, as well as fixed effects for the month the previous lease started by month the current lease started (to account for seasonal effects), as well as the number of months between leases. Standard errors double clustered at the unit and time level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.3: Dynamics of Price Adjustment: Data and Model

Panel A: Data						
	Share of Rent Changes			Size of Rent Changes		
	Increases	Decreases	Unchanged	Increases	Decreases	Combined
4Q Log Rent Index	3.432*** (0.213)	-2.761*** (0.204)	-0.671*** (0.074)	0.661*** (0.088)	-0.357*** (0.069)	1.246*** (0.085)
<i>N</i>	85217	85217	85217	54382	10354	85217
<i>R</i> ²	0.323	0.293	0.276	0.467	0.433	0.441
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.68	0.18	0.14	0.17	0.11	0.10

Panel B: Model						
	Share of Rent Changes			Size of Rent Changes		
	Increases	Decreases	Unchanged	Increases	Decreases	Combined
4Q Log Rent Index	3.379*** (0.199)	-2.572*** (0.181)	-0.807*** (0.052)	0.762*** (0.071)	-0.293*** (0.032)	1.174*** (0.088)
<i>N</i>	85217	85217	85217	55522	10229	85217
<i>R</i> ²	0.297	0.266	0.204	0.573	0.516	0.515
Unit FE	Yes	Yes	Yes	Yes	Yes	Yes
Dep. Mean	0.69	0.18	0.13	0.17	0.09	0.10

Notes: This table shows regressions in both the data and the model-simulated data estimating how rent changes respond to the business cycle. Panel (A) shows the estimates from the data, while Panel (B) shows the estimates from the model-generated data. Regressions include unit and months-since-last-lease fixed effects. Standard errors clustered by date in parentheses.

Table B.4: Berkeley New-Tenant Rent Inflation in Data and Model

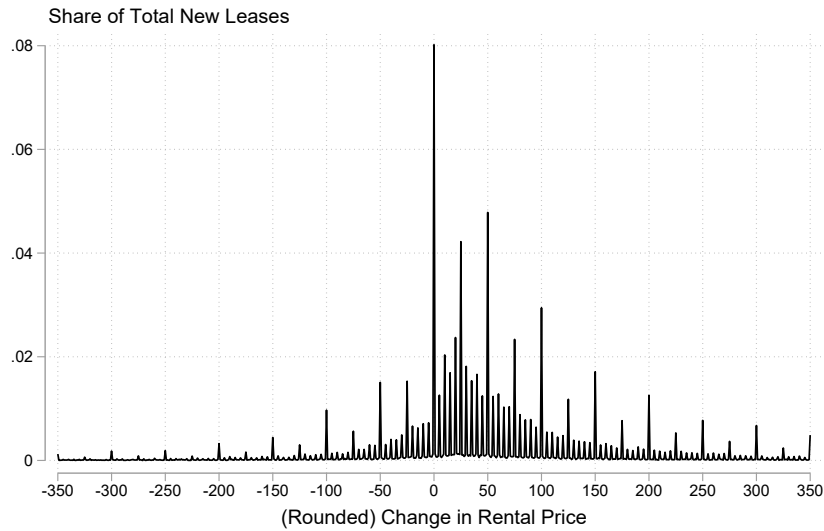
Series	All Years			Market Growth		
	Mean	SD	IQR	Low	Moderate	High
Data	4.024	5.107	7.996	-1.331	4.216	9.951
Model Baseline	4.059	5.056	7.848	-1.247	4.220	9.961
Double Large-Landlord Share	4.257	5.312	8.286	-1.281	4.357	10.486
All Large Landlords	4.455	5.563	8.705	-1.304	4.490	11.001

Notes: This table shows moments of the average inflation distribution in the data, our baseline model, and under our two counterfactuals.

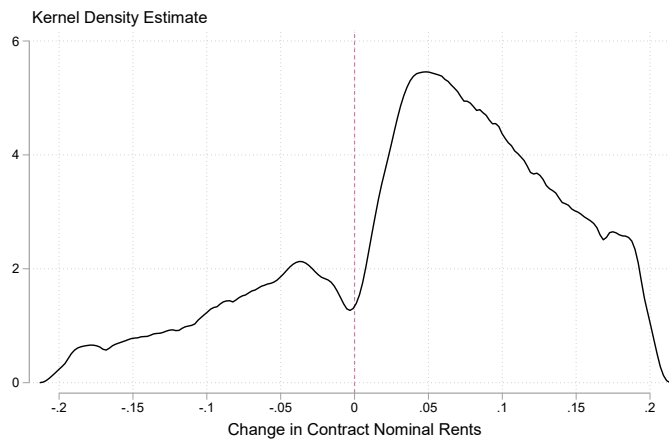
C More Evidence from the American Housing Survey

Figure C.1: Downward Nominal Rent Rigidity in the AHS

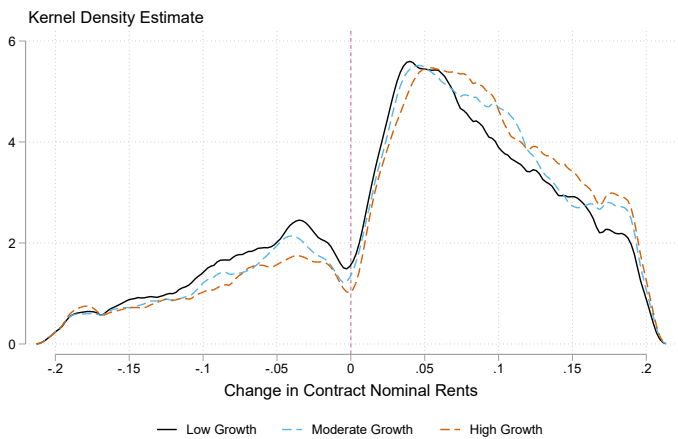
A. Overall Bunching at Zero Rent Changes



B. Kernel Density (excl. Zeros)



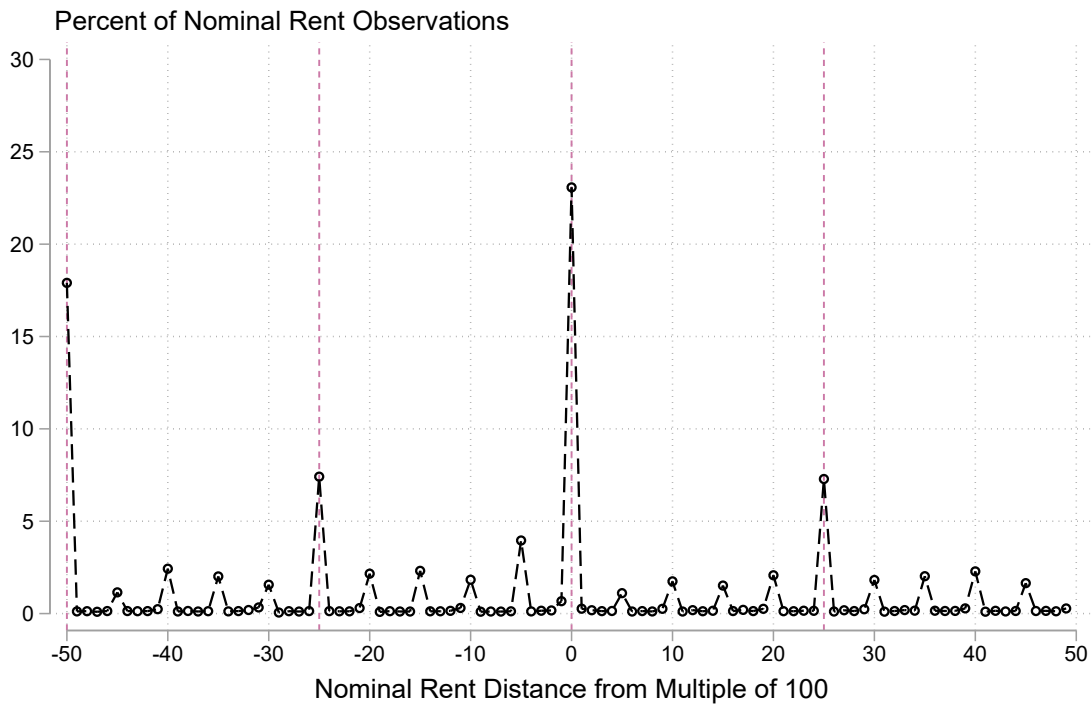
C. Kernel Densities by Rent Growth (excl. Zeros)



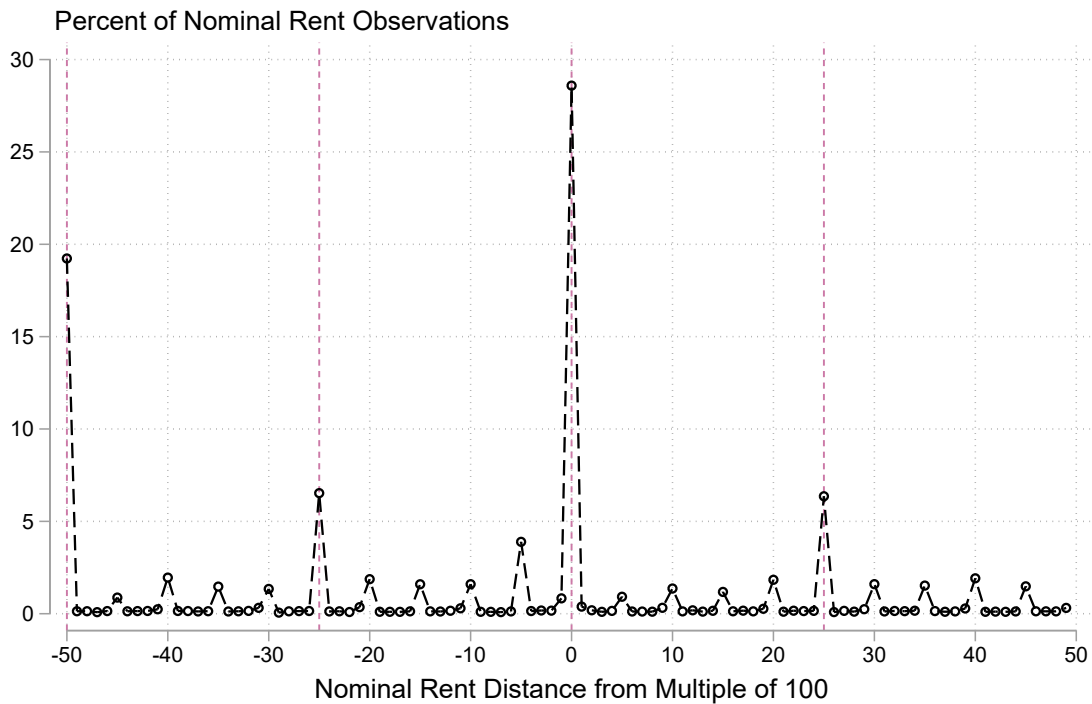
Notes: Panel (A) shows the histogram of the nominal dollar change in rents, rounded to the nearest dollar, in the AHS. We only include observations where the household differs from the household interviewed in the previous survey wave. Panel (B) shows the kernel density plots of the log change in contract rents, excluding zeros and dropping observations where the rent change is large at 20 log points. Panel (C) shows the same kernel density plots, but separately for low, moderate, and high rent growth periods.

Figure C.2: Heuristic Pricing in the Broader American Rental Market

A. 1985 - 2013 Sample



B. 1999 - 2013 Sample



Notes: This figure shows the percent of all observations in the AHS dataset with numbers based on their distance from the nearest multiple of \$100. Panel (A) shows the results for the period between 1985 and 2013, while Panel (B) shows the results for the period between 1999 and 2013. All counts are weighted using their sampling weights.

D Model Details

This section provides additional detail of the quantitative model presented in Section 5.

D.1 Information and Rent Target

Consider rental unit i , owned by a landlord of type $g \in \{S, L\}$, where S and L denote small and large landlords. The frictionless rent for a unit evolves due to both changes in aggregate rental-market conditions and idiosyncratic changes in demand for the individual unit. Let the full-information change in the frictionless log rent for unit i between a lease at time t and a lease at time $t - 1$ be given by:

$$\Delta m_{i,t} = M_{i,t} + U_{i,t} \quad (4)$$

where $M_{i,t}$ is the innovation due to changes in market conditions and $U_{i,t}$ is the unit-specific innovation. We assume that both components are mean zero and accumulate with the length between leases:

$$M_{i,t} \sim N(0, \sigma_M^2 L_{i,t}), \quad U_{i,t} \sim N(0, \sigma_U^2 L_{i,t})$$

Landlords of type g observe noisy signals of each component. The signals for each component are given by:

$$S_{i,t,g}^M = M_{i,t} + \nu_{i,t,g}^M, \quad S_{i,t,g}^U = U_{i,t} + \nu_{i,t,g}^U.$$

Similarly, we assume that the noise is mean zero and accumulates with the lease length:

$$\nu_{i,t,g}^M \sim N(0, \sigma_{\nu^M,g}^2 L_{i,t}), \quad \nu_{i,t,g}^U \sim N(0, \sigma_{\nu^U,g}^2 L_{i,t})$$

Then, the landlord's posterior is given by:

$$\widetilde{\Delta m}_{i,t,g} = K_g^M S_{i,t,g}^M + K_g^U S_{i,t,g}^U$$

Where $K_g^M = \frac{\sigma_M^2}{\sigma_M^2 + \sigma_{\nu^M,g}^2}$ and $K_g^U = \frac{\sigma_U^2}{\sigma_U^2 + \sigma_{\nu^U,g}^2}$ are the Kalman gains.

In estimation, we use our estimated repeat-rent index as a proxy for the market innovation, and assume that it summarizes the market-wide information that landlords use when setting prices. We assume the cumulative change in the repeat-rent index is related to the market innovation through the following functional form:

$$A_{i,t} = \omega M_{i,t} + \zeta_{i,t}$$

Then, because the repeat-rent index and landlords' posteriors are jointly normal, the pos-

terior of the market component is given by: $\theta_g A_t$. Finally, because the unit-specific component has variance given by the following expression:

$$\text{Var}(S_{i,t,g}^U | L_{i,t}) = (\sigma_U^2 + \sigma_{v^U,g}^2) L_{i,t},$$

Then, the posterior mean has the following variance:

$$\text{Var}(\tilde{U}_{i,t,g} | L_{i,t}) = \frac{\sigma_U^4}{\sigma_U^2 + \sigma_{v^U,g}^2} L_{i,t}$$

Then, defining $\sigma_{z,g}^2 = \frac{\sigma_U^4}{\sigma_U^2 + \sigma_{v^U,g}^2}$, gives the following expression for the posterior of the unit-specific component, where $\epsilon_{i,t} \sim N(0, 1)$:

$$\tilde{U}_{i,t,g} = \sigma_{z,g} \sqrt{L_{i,t}} \epsilon_{i,t},$$

Combining these together, the landlord's desired rent adjustment is given by:

$$x_{i,t,g} = b_{i,t-1,g} + \theta_g A_t + \sigma_{z,g} \sqrt{L_{i,t}} \epsilon_{i,t}.$$

where $b_{i,t-1,g}$ captures any pricing gap inherited from a previous reset opportunity. In estimation, we assume that the first observed gap for each unit has $b_{i,t-1,g} = 0$. When a landlord reoptimizes, they set their rent to the desired rent, and thus the incumbent pricing gap at the next opportunity to reset their rent is set to zero.

D.2 Adjustment Costs and Reoptimization Decision

At turnover, the landlord draws a reoptimization cost, which takes the following form (Blanco et al., 2025):

$$K_{i,t,g}^r = \begin{cases} 0, & \text{with probability } 1 - \lambda_g^r, \\ \kappa_g^r U_{i,t}^r, & \text{with probability } \lambda_g^r, \end{cases} \quad U_{i,t}^r \sim U[0, 1] \quad (5)$$

In addition, landlords draw an additional cost that they pay when reducing rents below their previous value. This rent-cut cost takes the following form:

$$K_{i,t,g}^c = \kappa_g^c U_{i,t}^c, \quad U_{i,t}^c \sim U[0, 1] \quad (6)$$

When choosing whether or not to reoptimize, landlords weigh these fixed costs against the cost of mispricing. The loss from mispricing is approximately quadratic in the gap between the

optimal price and incumbent price.³⁶ Thus, the landlord chooses to reoptimize and set prices at the desired rent when:

$$\frac{1}{2}x_{i,t,g}^2 \geq K_{i,t,g}^r + \mathbf{1}\{x_{i,t,g} < 0\}K_{i,t,g}^c.$$

³⁶We normalize, without loss of generality, the curvature parameter to 1.